

Blockchain Technologies: an Introduction

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The Hipe

Blockchain technology: the future of the banking sector

Blockchain Development

Blockchain Technology Market to Reach USD 7.74 Billion By 2024

Jan 20, 2017 6:10 PM EST by Jessie Willms

Report: Blockchain Technology Market to Reach \$7.7 Billion by 2024

10 ottobre 2017, 06:01 CEST

Welcome to Crypto Valley

The Swiss city of Zug embraces digital currency. Some worry that the money might be a little too secret.

Blog

11 January, 2017

The importance of the blockchain: The second revolution of the internet

iammond

omments

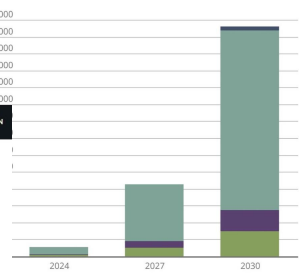
Pagare casa in Bitcoin, primo caso in Italia

Una giovane cinese ha acquistato un appartamento in una zona centrale di Torino, pagando in Bitcoin.

di Alessandro Crea - 29 Gennaio 2018, 19:30 - (Fonte AGI)



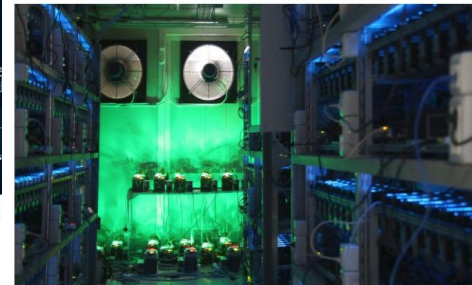
Annual Bank Savings from Blockchain Implementation in Cross-border B2B Settlement (\$m)



LA STAMPA ECONOMIA

Il bitcoin potrebbe diventare l'arma segreta dell'antipolitica

Una tecnologia in grado di riuscire dove i sistemi tradizionali falliscono



f 151 t G+ in

L'anti-politica afferma di rappresentare la gente comune, contro il sistema. Le anti-valute sono definite (dalla Bce) monete accettate dalla gente comune, non emesse dal sistema. Due sfide convergenti, oppure due raggi? Bitcoin, l'anti-valuta più nota, è l'arma segreta dell'anti-politica, oppure uno strumento di speculazione?

Negozi CFD su Ethereum, Litecoin

Il tuo capitale potrebbe essere a rischio

Scopri di più

tariffa.it

L'app di Tariffa.it ti suggerisce la tariffa telefonica mobile che ti conviene di più. L'app è semplice, precisa! Scarica il tuo smartphone Android!

VIDEO CONSIGLIATI

L'app di Tariffa.it ti suggerisce la tariffa telefonica mobile che ti conviene di più. L'app è semplice, precisa! Scarica il tuo smartphone Android!

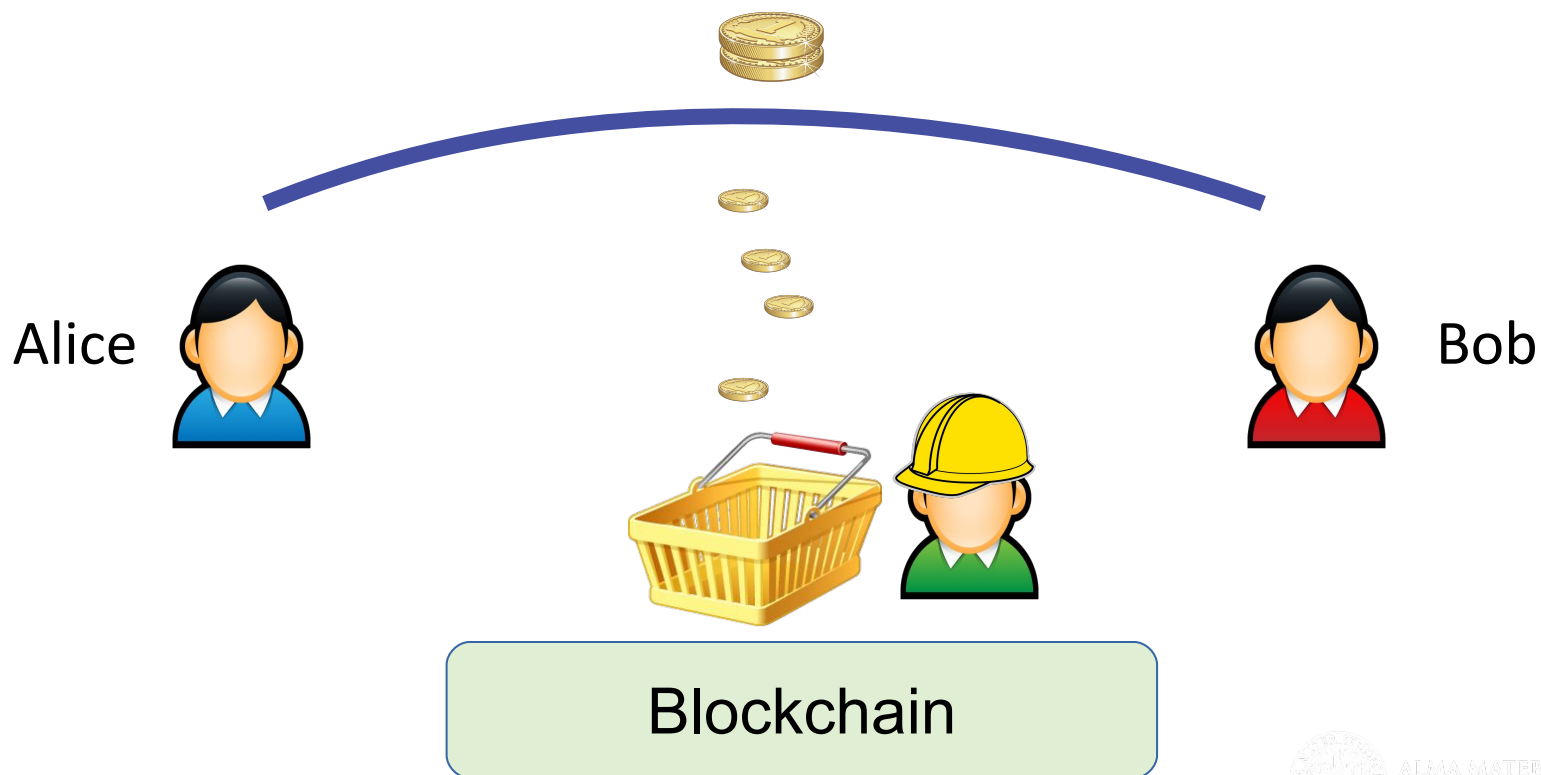
NOTIZIE / PREVISIONI / APPROFONDIMENTI / CRIPTOVALUTE EMERGENTI /

LAPEI STA STUDIANDO DI USARE IOTA PER IDENTIFICAZIONE DEI CITTADINI (LE CARTE DI IDENTITÀ)

novazione di Marco Cavicchioli 9 febbraio 2018



Blockchain?



Blockchain?



There's more?

Al

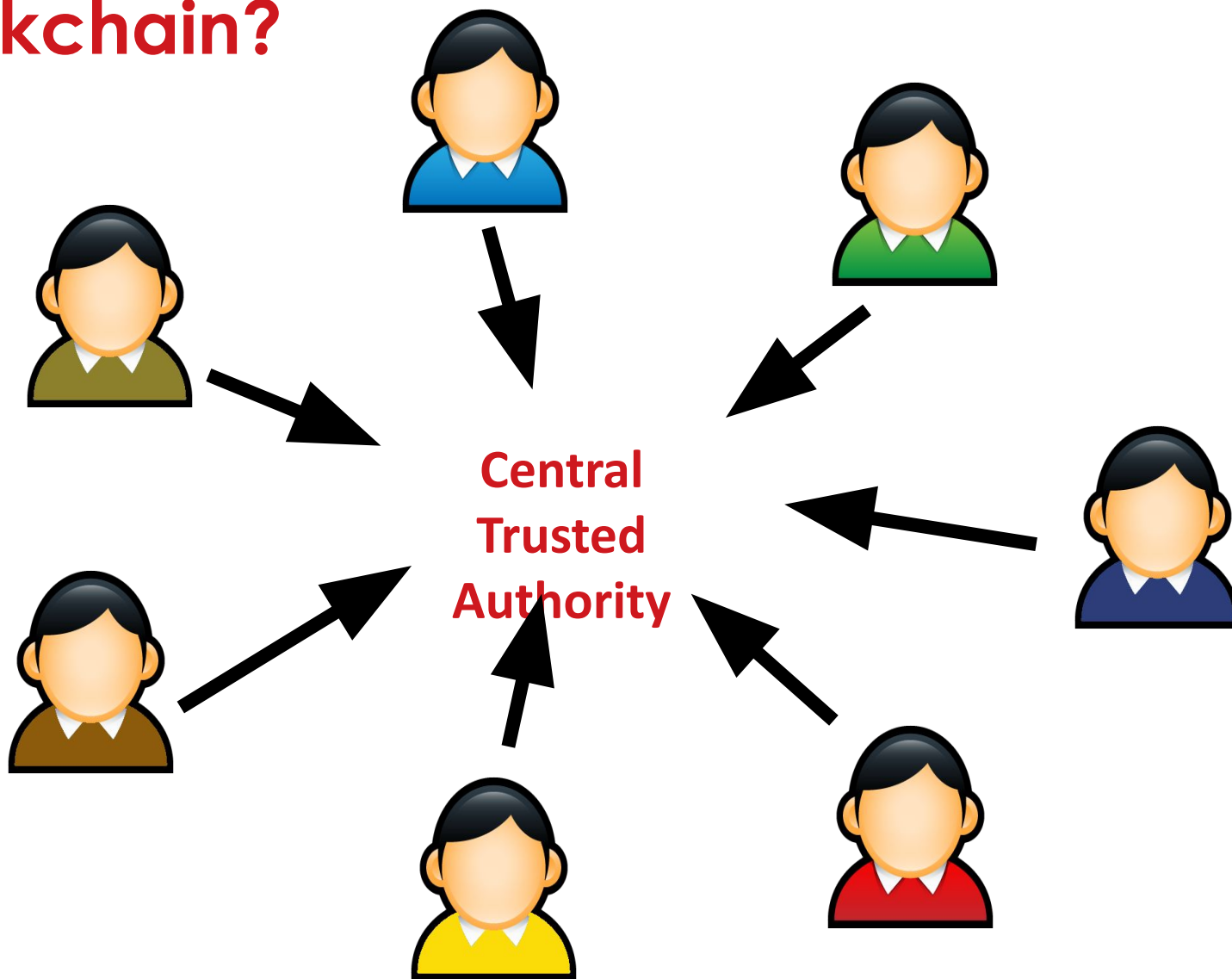


Blockchain

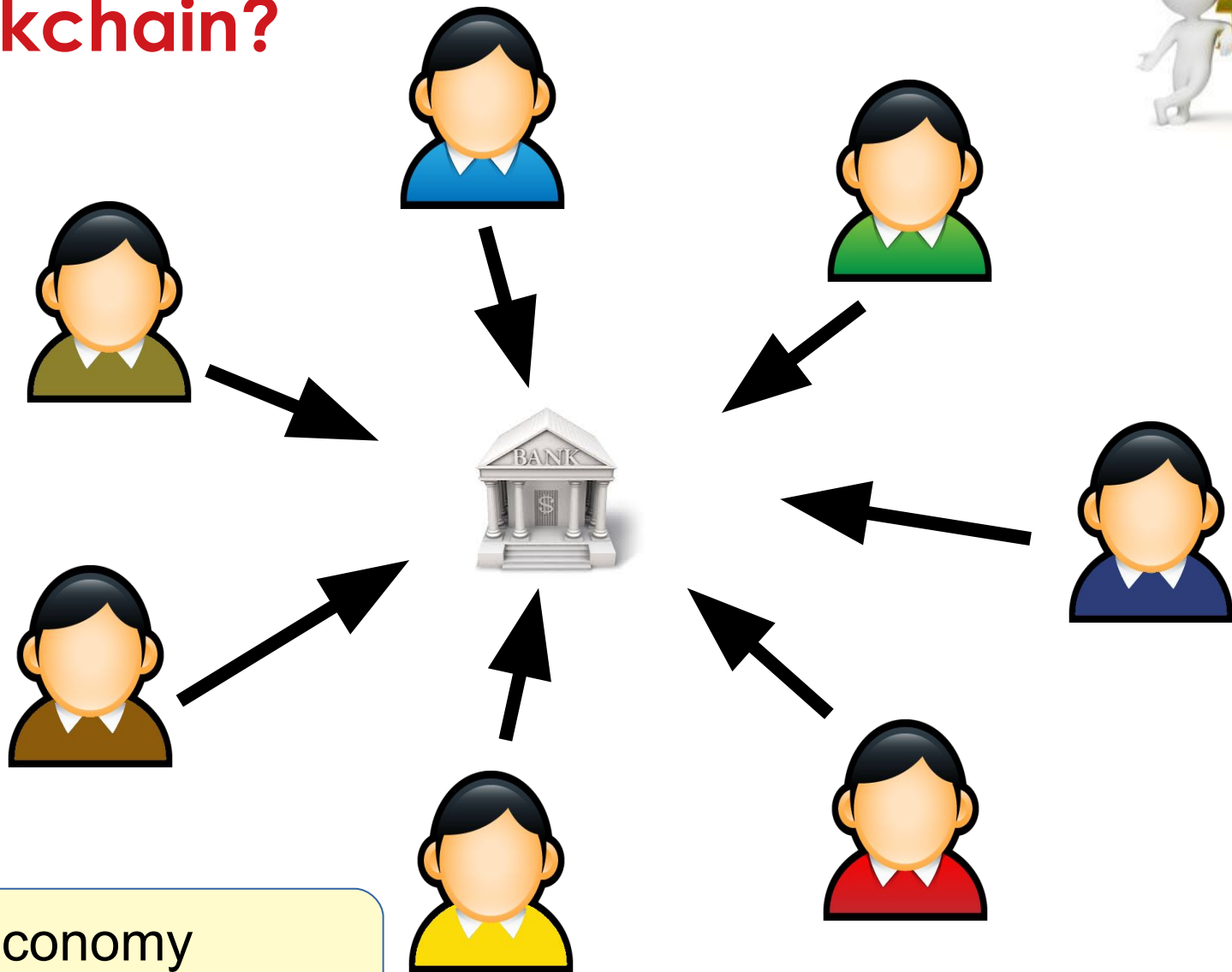


ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Blockchain?



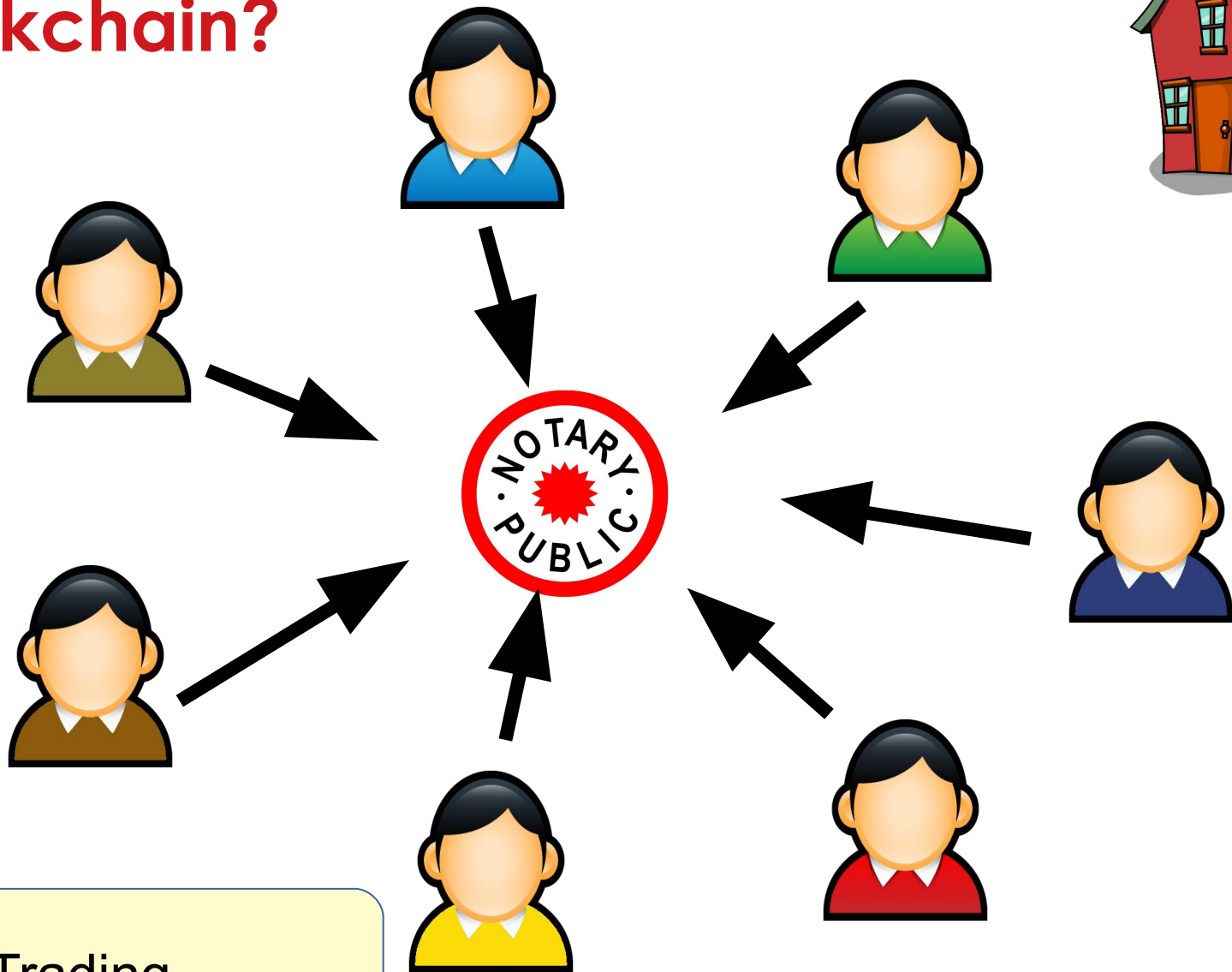
Blockchain?



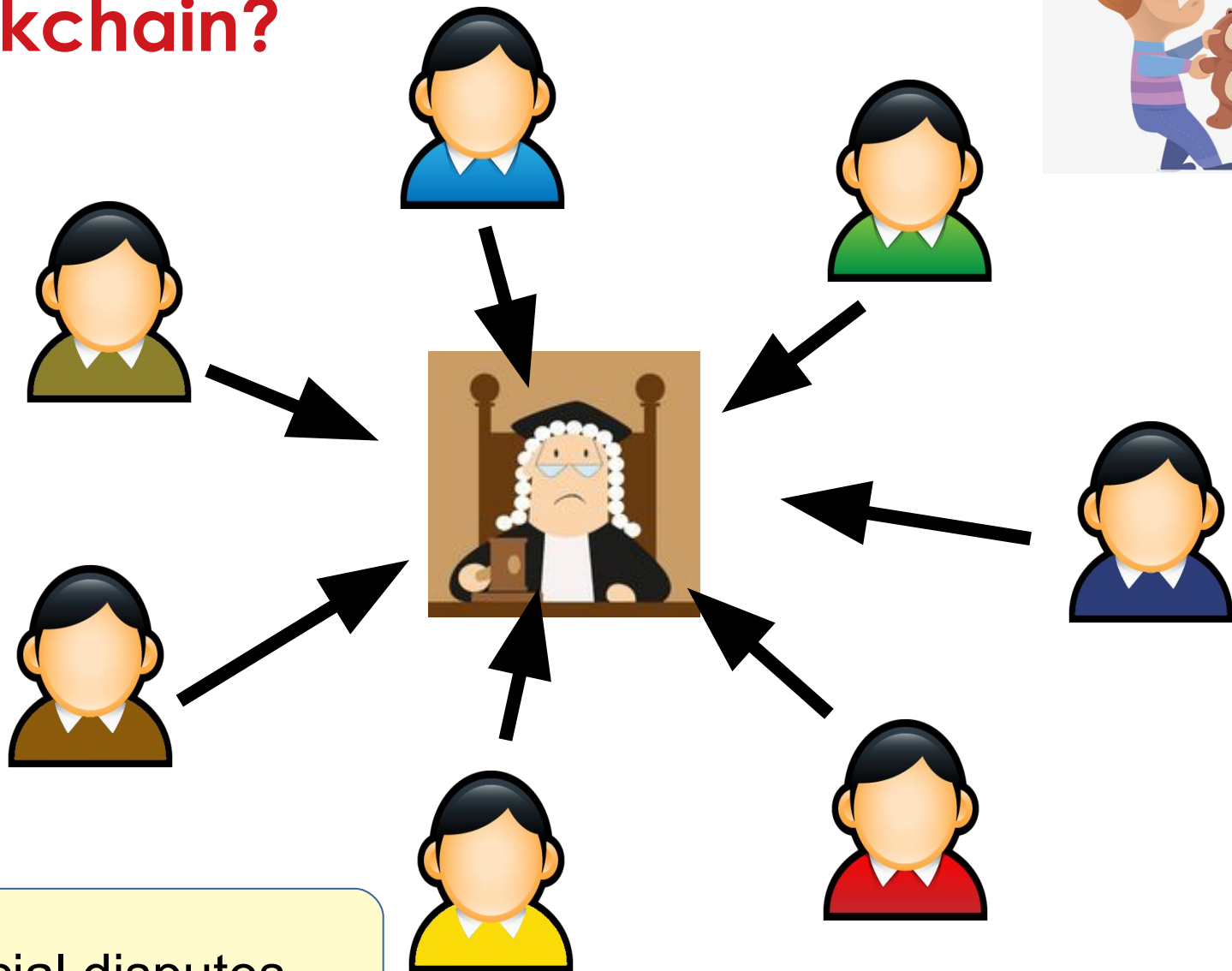
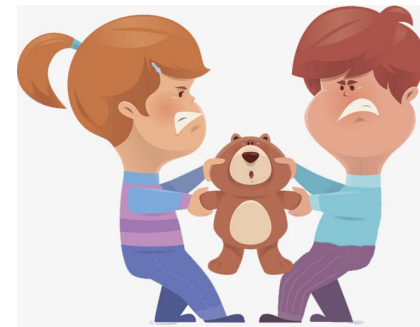
Economy
Finance



Blockchain?



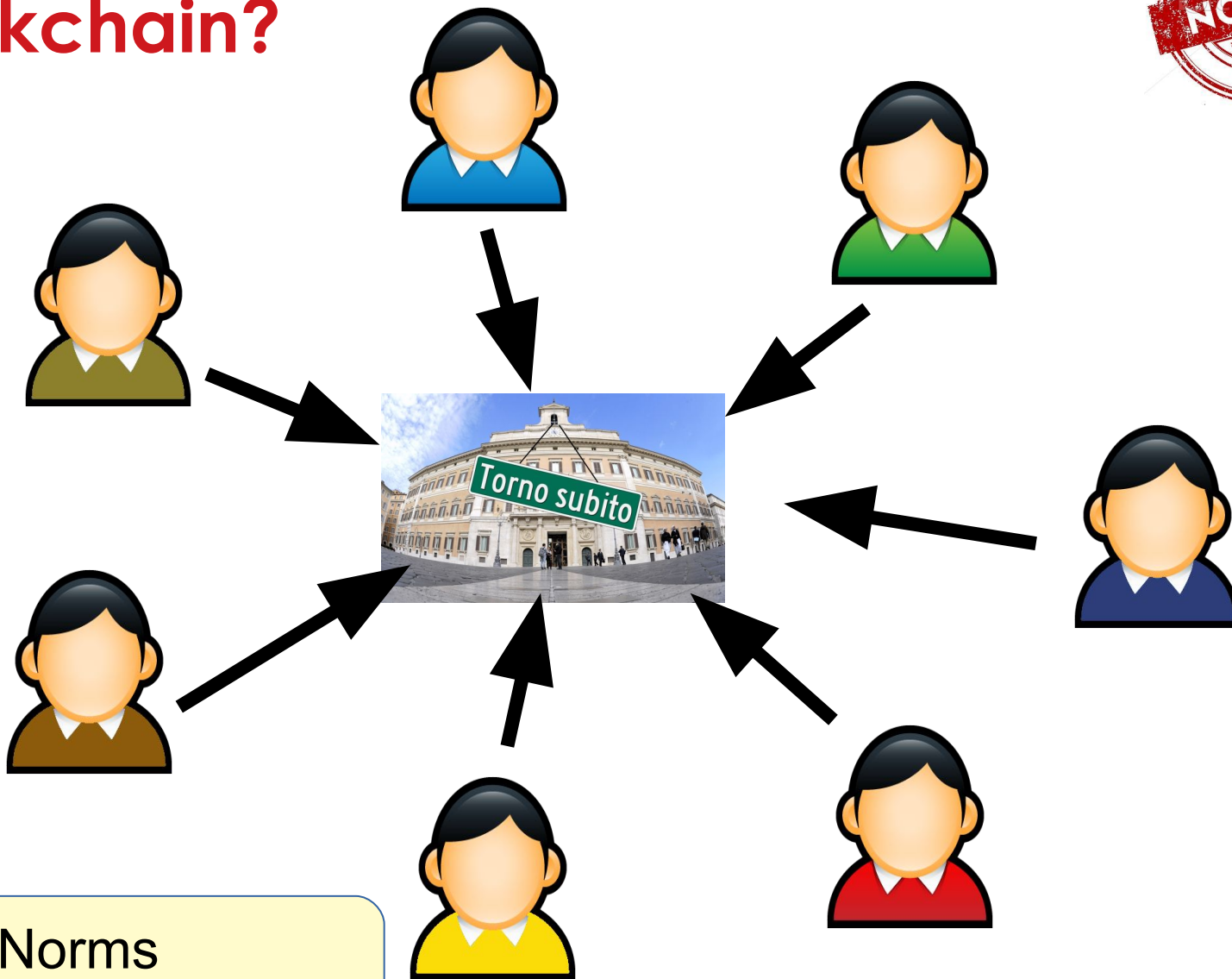
Blockchain?



Jiudicial disputes



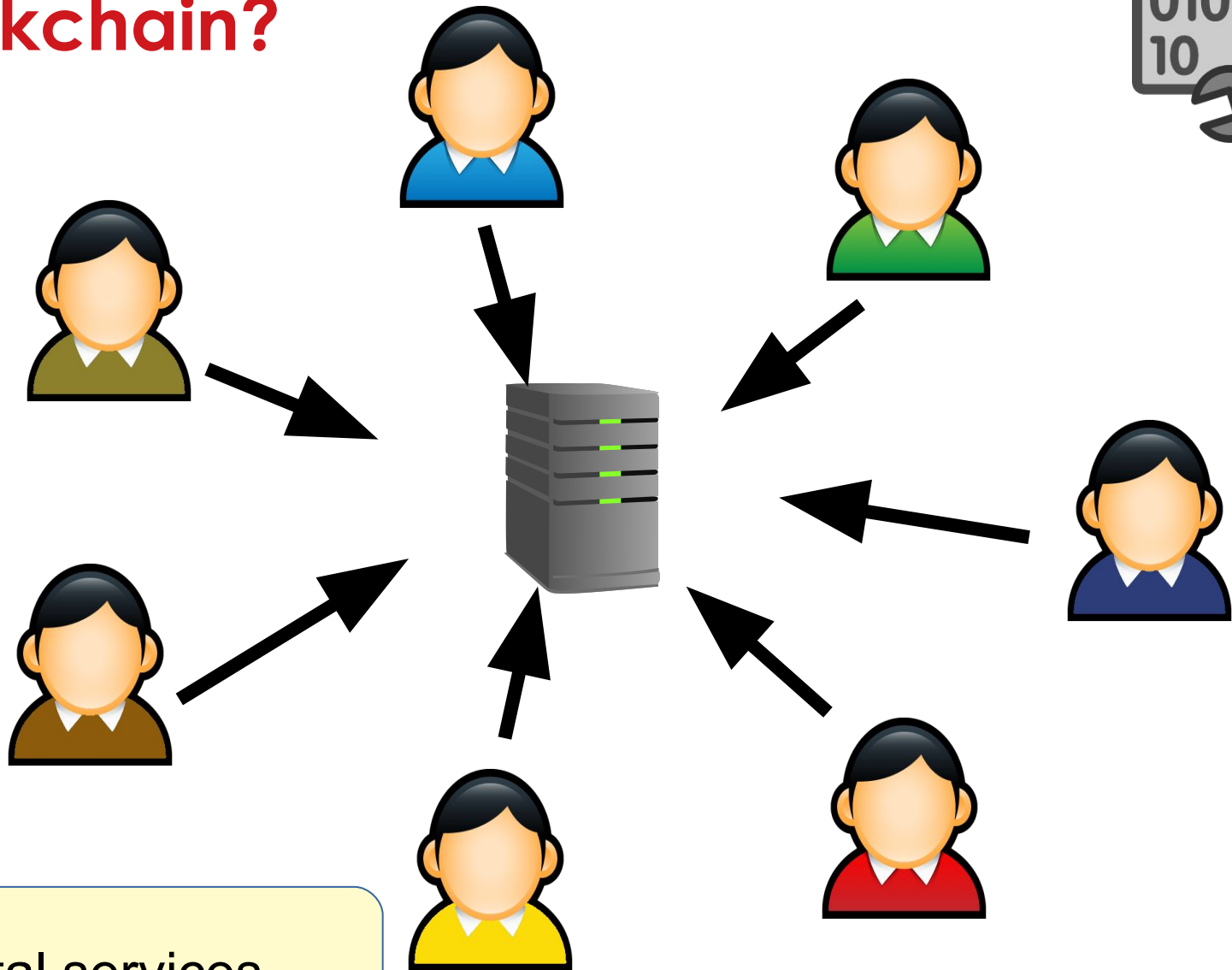
Blockchain?



Norms
Regulation



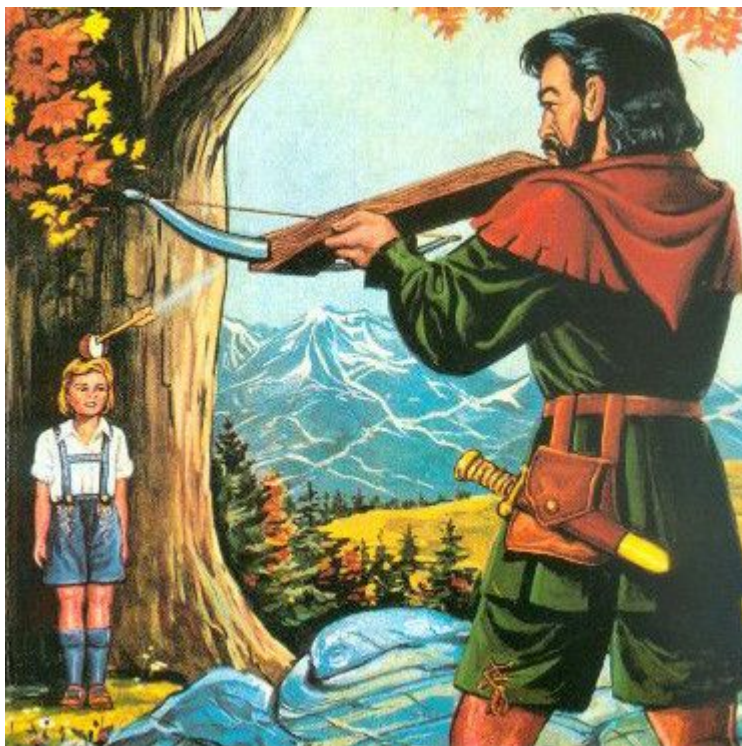
Blockchain?



Digital services



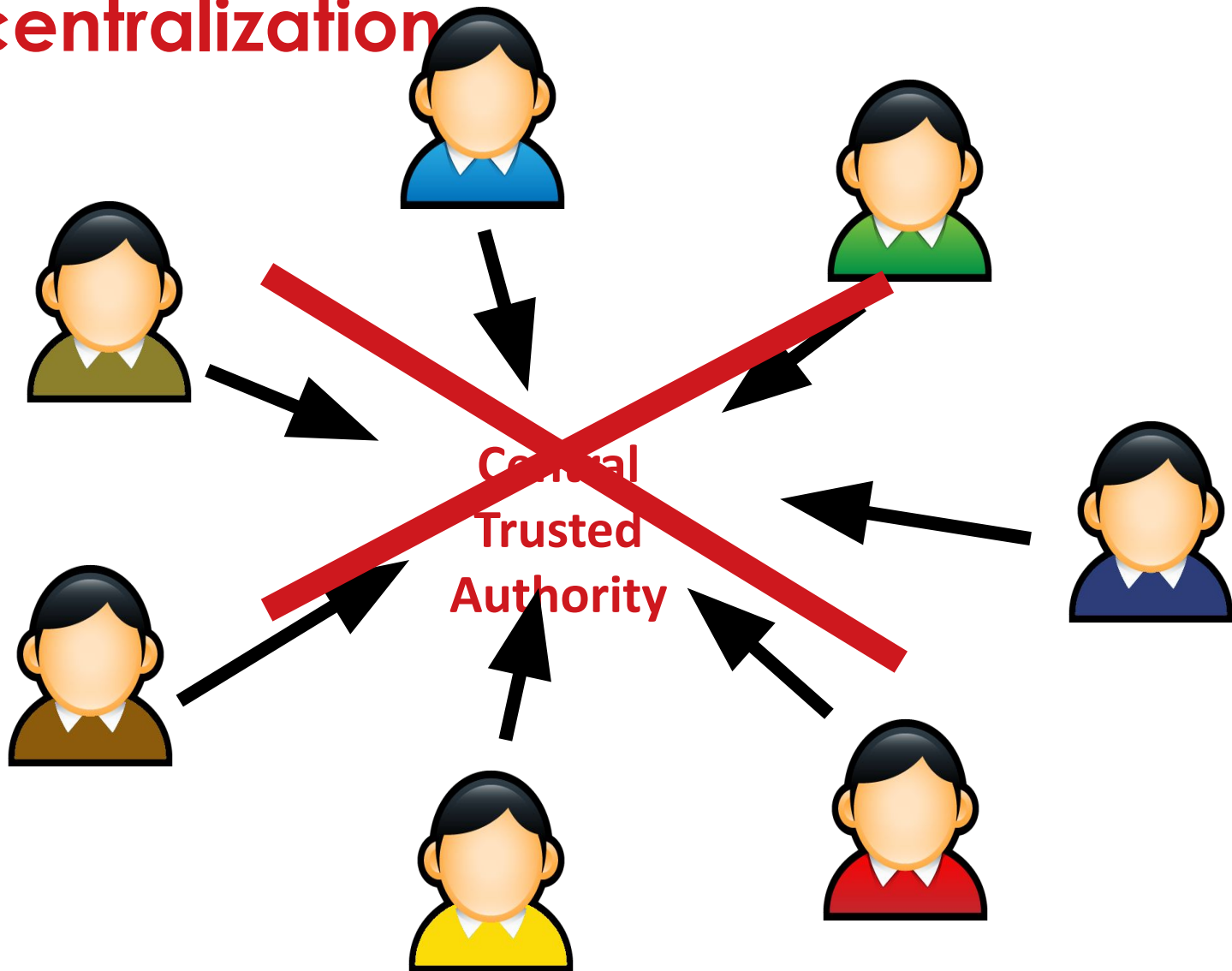
Central Trusted Authority



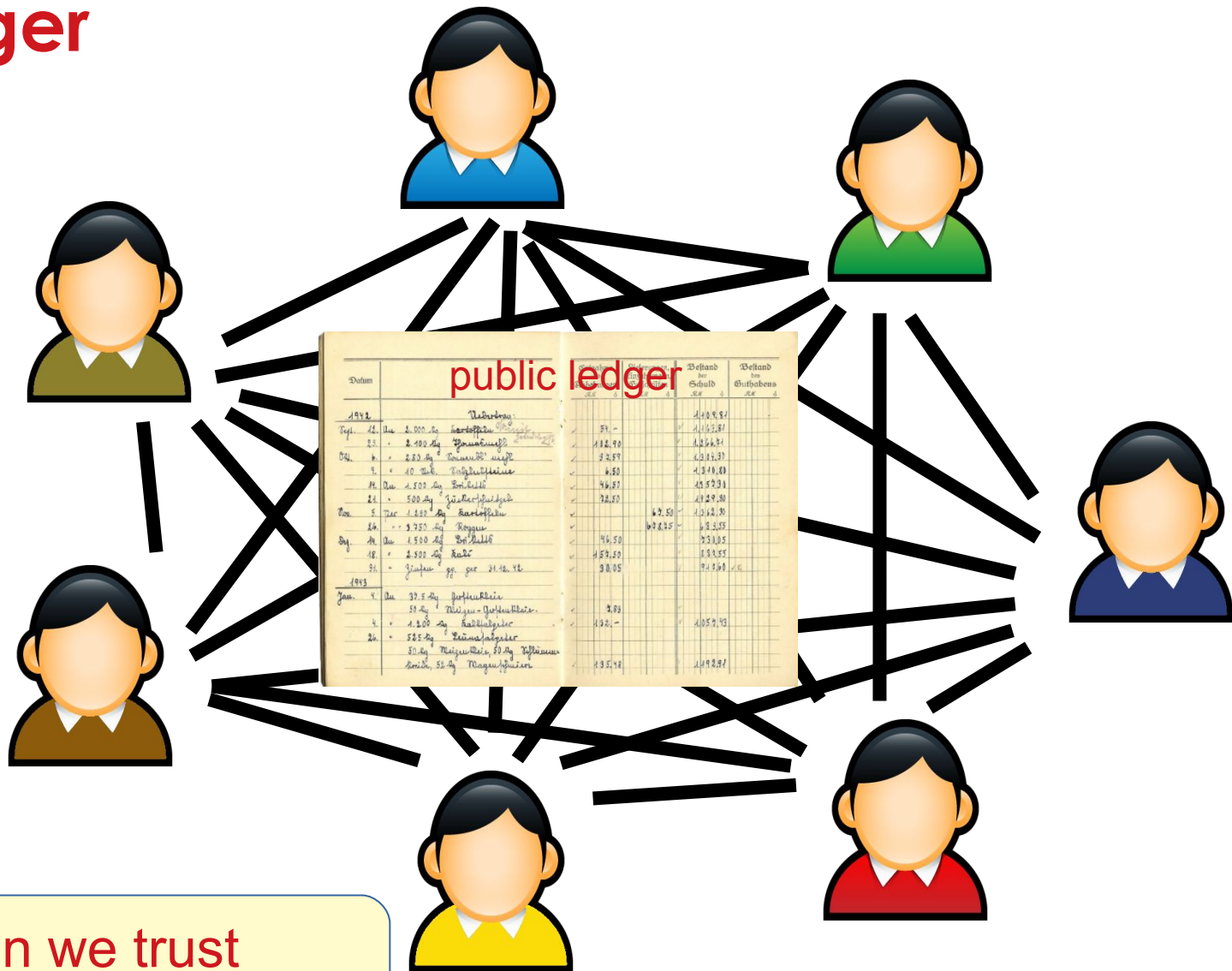
Trust



No centralization



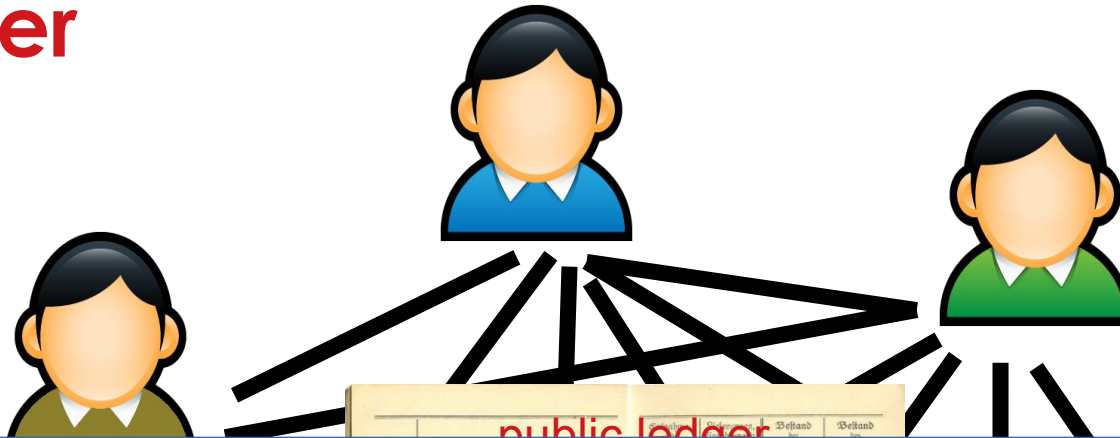
Ledger



Can we trust
the others?

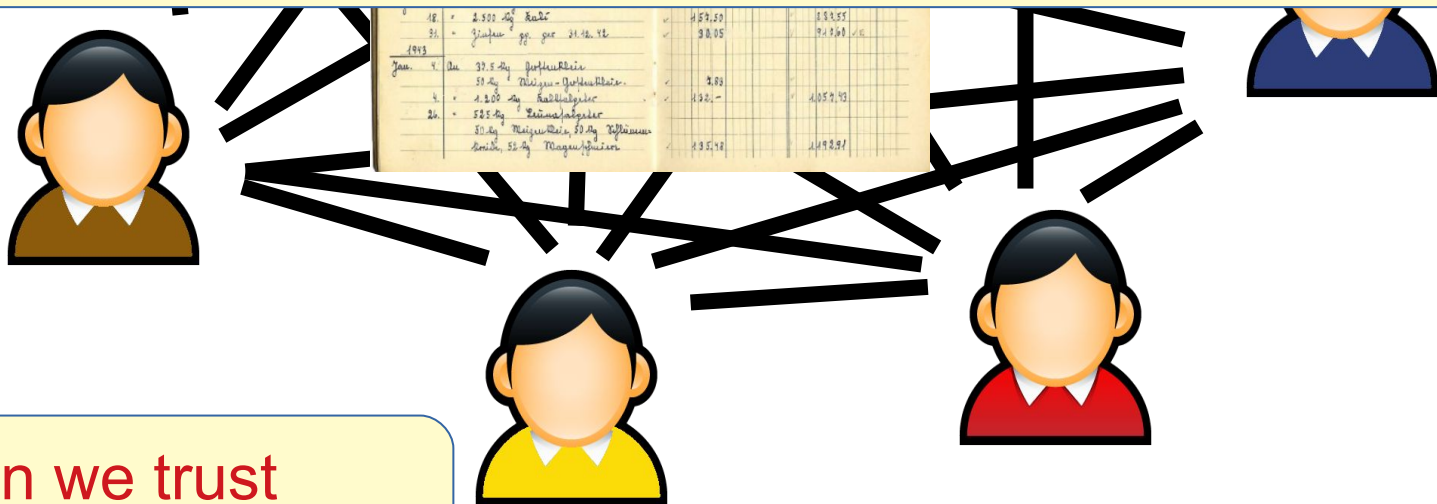


Ledger



Desiderata:

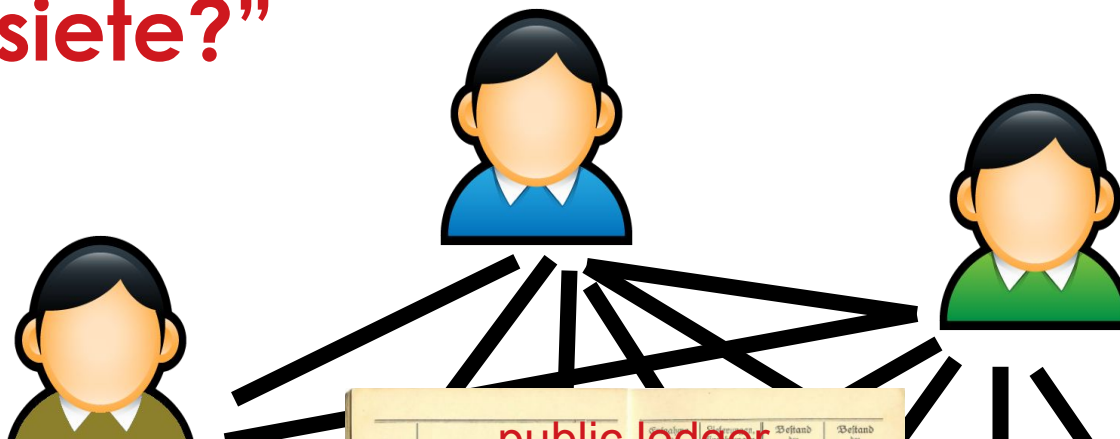
Users are able to build **trust** *without* involving a third party



Can we trust
the others?

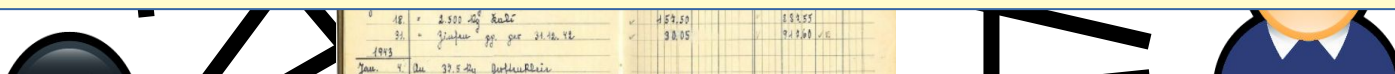


“Chi siete?”



Desiderata:

Users are able to build **trust** *without* involving a third party



Ex: can we build

Uber without Uber, or Airbnb without Airbnb?



Can we trust
the others?



do-Anonymization

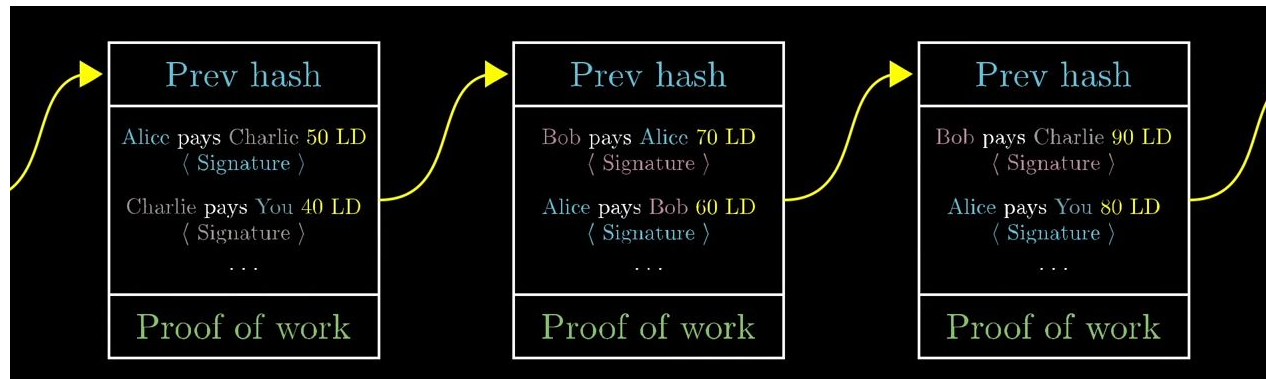
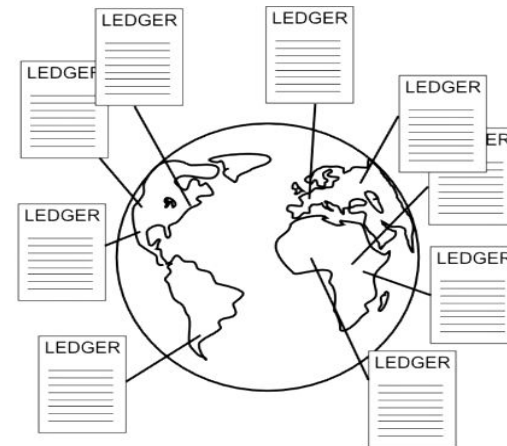
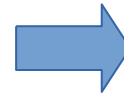
The diagram shows a central 'public ledger' (a ledger book) connected to eight anonymous participants (represented by masked figures). Each participant holds a private key (S) and a public key (P). The ledger contains a list of transactions, including:

Datum			Bestand der Schuld	Bestand des Guthabens
1912				
Febr. 21.	An 2.000.-	Karlsruhe	31.-	1.009,51
22.	An 2.100.-	Hannover	112,90	1.006,61
März 1.	An 2.200.-	Karlsruhe	32,57	1.003,03
9.	An 10.000.-	Karlsruhe	4,50	1.000,00
14.	An 1.500.-	Karlsruhe	96,57	1.000,00
24.	An 500.-	Karlsruhe	92,57	1.000,00
25.	An 1.500.-	Karlsruhe	4,50	1.000,00
26.	An 1.500.-	Karlsruhe	96,57	1.000,00
27.	An 1.500.-	Karlsruhe	96,57	1.000,00
28.	An 1.500.-	Karlsruhe	96,57	1.000,00
29.	An 1.500.-	Karlsruhe	96,57	1.000,00
30.	An 1.500.-	Karlsruhe	96,57	1.000,00
31.	An 1.500.-	Karlsruhe	96,57	1.000,00
1913				
Jan. 1.	An 37,50	Karlsruhe	31,03	1.000,00
1.	An 1.500.-	Karlsruhe	112,90	1.000,00
21.	An 1.500.-	Karlsruhe	112,90	1.000,00
31.	An 1.500.-	Karlsruhe	112,90	1.000,00

Below the ledger, a yellow box contains the text: "n we trust e others?"

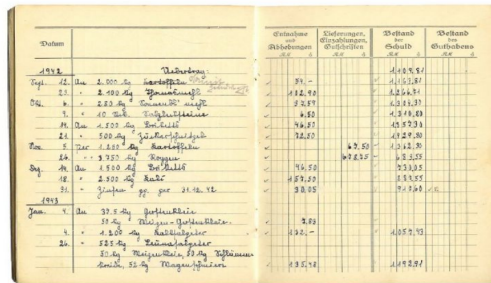
Can we trust the others?

Redundancy

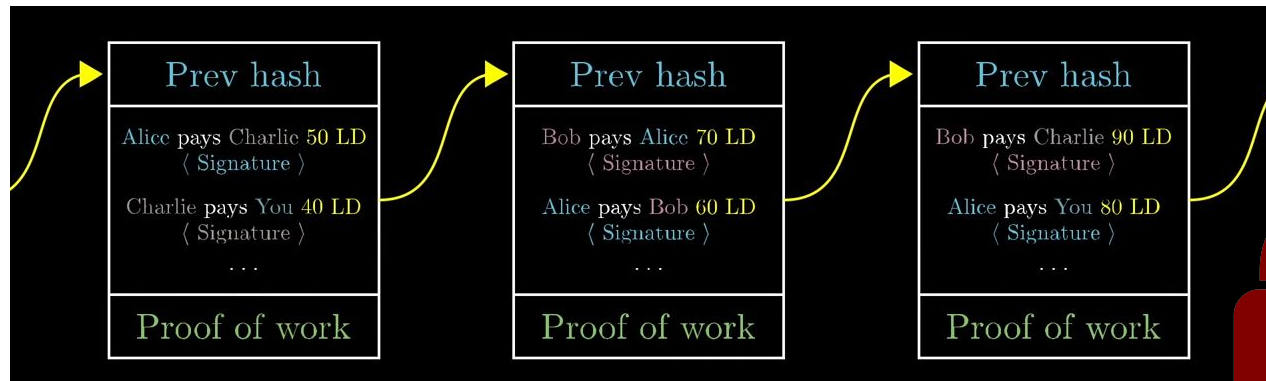
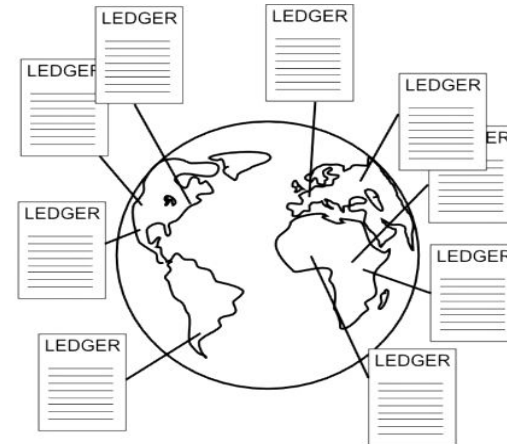
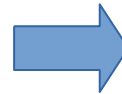
[illegible]

append-only log

Redundancy + Untamperability



Datum	Bezeichnung	Einnahme oder Abgaben	Ausgaben, Entlohnungen, Schulden	Stellend des Saldo	Stellend des Guthabens
1718	1. Jan. 1. 100. -	100. -		100. -	
1719	1. Jan. 1. 100. -	100. -		100. -	
1720	1. Jan. 1. 100. -	100. -		100. -	
1721	1. Jan. 1. 100. -	100. -		100. -	
1722	1. Jan. 1. 100. -	100. -		100. -	
1723	1. Jan. 1. 100. -	100. -		100. -	
1724	1. Jan. 1. 100. -	100. -		100. -	
1725	1. Jan. 1. 100. -	100. -		100. -	
1726	1. Jan. 1. 100. -	100. -		100. -	
1727	1. Jan. 1. 100. -	100. -		100. -	
1728	1. Jan. 1. 100. -	100. -		100. -	
1729	1. Jan. 1. 100. -	100. -		100. -	
1730	1. Jan. 1. 100. -	100. -		100. -	
1731	1. Jan. 1. 100. -	100. -		100. -	
1732	1. Jan. 1. 100. -	100. -		100. -	
1733	1. Jan. 1. 100. -	100. -		100. -	
1734	1. Jan. 1. 100. -	100. -		100. -	
1735	1. Jan. 1. 100. -	100. -		100. -	
1736	1. Jan. 1. 100. -	100. -		100. -	
1737	1. Jan. 1. 100. -	100. -		100. -	
1738	1. Jan. 1. 100. -	100. -		100. -	
1739	1. Jan. 1. 100. -	100. -		100. -	
1740	1. Jan. 1. 100. -	100. -		100. -	
1741	1. Jan. 1. 100. -	100. -		100. -	
1742	1. Jan. 1. 100. -	100. -		100. -	
1743	1. Jan. 1. 100. -	100. -		100. -	
1744	1. Jan. 1. 100. -	100. -		100. -	
1745	1. Jan. 1. 100. -	100. -		100. -	
1746	1. Jan. 1. 100. -	100. -		100. -	
1747	1. Jan. 1. 100. -	100. -		100. -	
1748	1. Jan. 1. 100. -	100. -		100. -	
1749	1. Jan. 1. 100. -	100. -		100. -	
1750	1. Jan. 1. 100. -	100. -		100. -	
1751	1. Jan. 1. 100. -	100. -		100. -	
1752	1. Jan. 1. 100. -	100. -		100. -	
1753	1. Jan. 1. 100. -	100. -		100. -	
1754	1. Jan. 1. 100. -	100. -		100. -	
1755	1. Jan. 1. 100. -	100. -		100. -	
1756	1. Jan. 1. 100. -	100. -		100. -	
1757	1. Jan. 1. 100. -	100. -		100. -	
1758	1. Jan. 1. 100. -	100. -		100. -	
1759	1. Jan. 1. 100. -	100. -		100. -	
1760	1. Jan. 1. 100. -	100. -		100. -	
1761	1. Jan. 1. 100. -	100. -		100. -	
1762	1. Jan. 1. 100. -	100. -		100. -	
1763	1. Jan. 1. 100. -	100. -		100. -	
1764	1. Jan. 1. 100. -	100. -		100. -	
1765	1. Jan. 1. 100. -	100. -		100. -	
1766	1. Jan. 1. 100. -	100. -		100. -	
1767	1. Jan. 1. 100. -	100. -		100. -	
1768	1. Jan. 1. 100. -	100. -		100. -	
1769	1. Jan. 1. 100. -	100. -		100. -	
1770	1. Jan. 1. 100. -	100. -		100. -	
1771	1. Jan. 1. 100. -	100. -		100. -	
1772	1. Jan. 1. 100. -	100. -		100. -	
1773	1. Jan. 1. 100. -	100. -		100. -	
1774	1. Jan. 1. 100. -	100. -		100. -	
1775	1. Jan. 1. 100. -	100. -		100. -	
1776	1. Jan. 1. 100. -	100. -		100. -	
1777	1. Jan. 1. 100. -	100. -		100. -	
1778	1. Jan. 1. 100. -	100. -		100. -	
1779	1. Jan. 1. 100. -	100. -		100. -	
1780	1. Jan. 1. 100. -	100. -		100. -	
1781	1. Jan. 1. 100. -	100. -		100. -	
1782	1. Jan. 1. 100. -	100. -		100. -	
1783	1. Jan. 1. 100. -	100. -		100. -	
1784	1. Jan. 1. 100. -	100. -		100. -	
1785	1. Jan. 1. 100. -	100. -		100. -	
1786	1. Jan. 1. 100. -	100. -		100. -	
1787	1. Jan. 1. 100. -	100. -		100. -	
1788	1. Jan. 1. 100. -	100. -		100. -	
1789	1. Jan. 1. 100. -	100. -		100. -	
1790	1. Jan. 1. 100. -	100. -		100. -	
1791	1. Jan. 1. 100. -	100. -		100. -	
1792	1. Jan. 1. 100. -	100. -		100. -	
1793	1. Jan. 1. 100. -	100. -		100. -	
1794	1. Jan. 1. 100. -	100. -		100. -	
1795	1. Jan. 1. 100. -	100. -		100. -	
1796	1. Jan. 1. 100. -	100. -		100. -	
1797	1. Jan. 1. 100. -	100. -		100. -	
1798	1. Jan. 1. 100. -	100. -		100. -	
1799	1. Jan. 1. 100. -	100. -		100. -	
1800	1. Jan. 1. 100. -	100. -		100. -	



append-only log

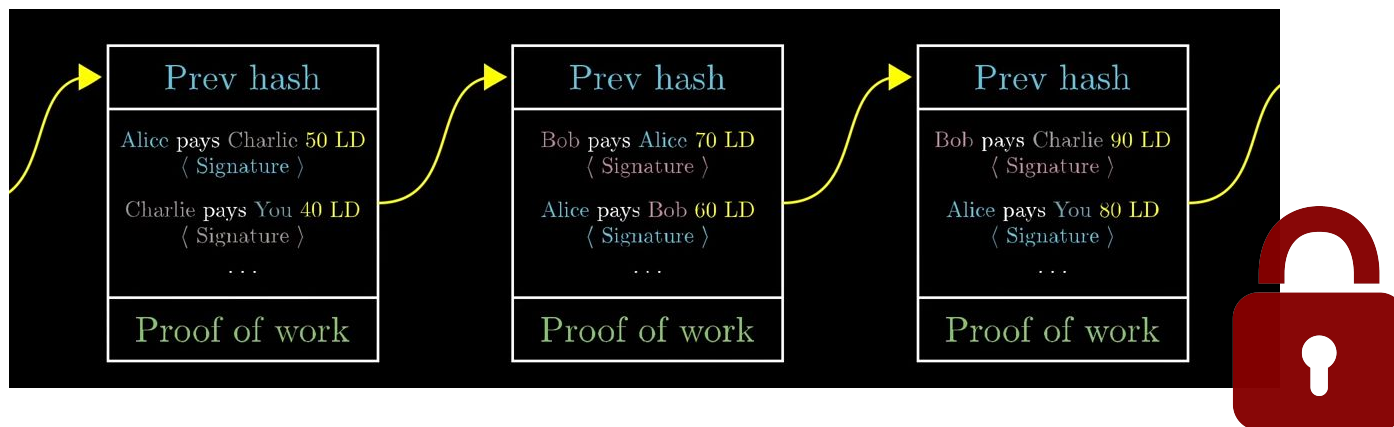


“Chi siete?”

Authenticity

Verifiability

Immutability



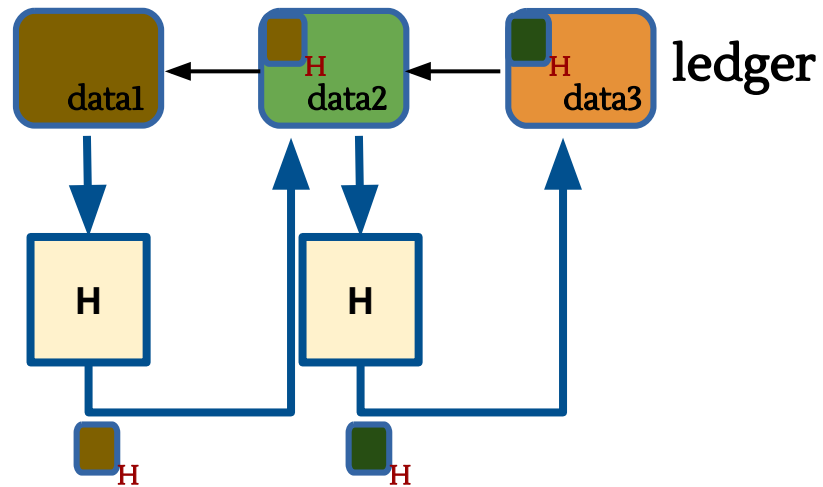
append-only log



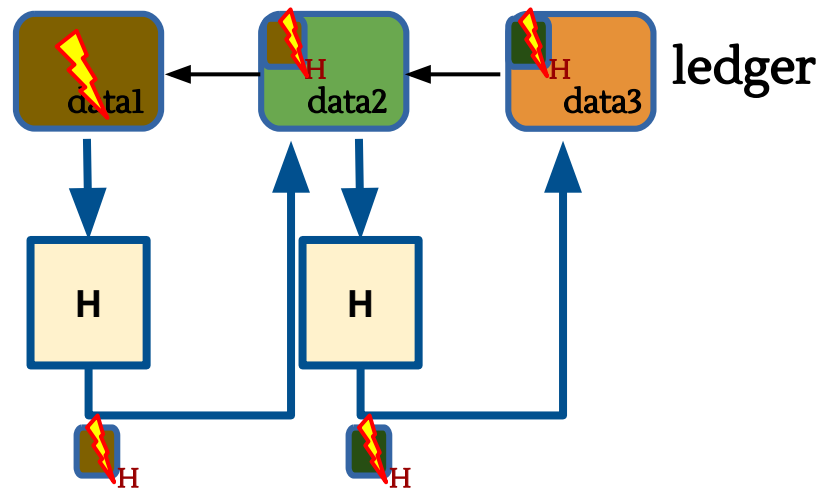
The Public Ledger



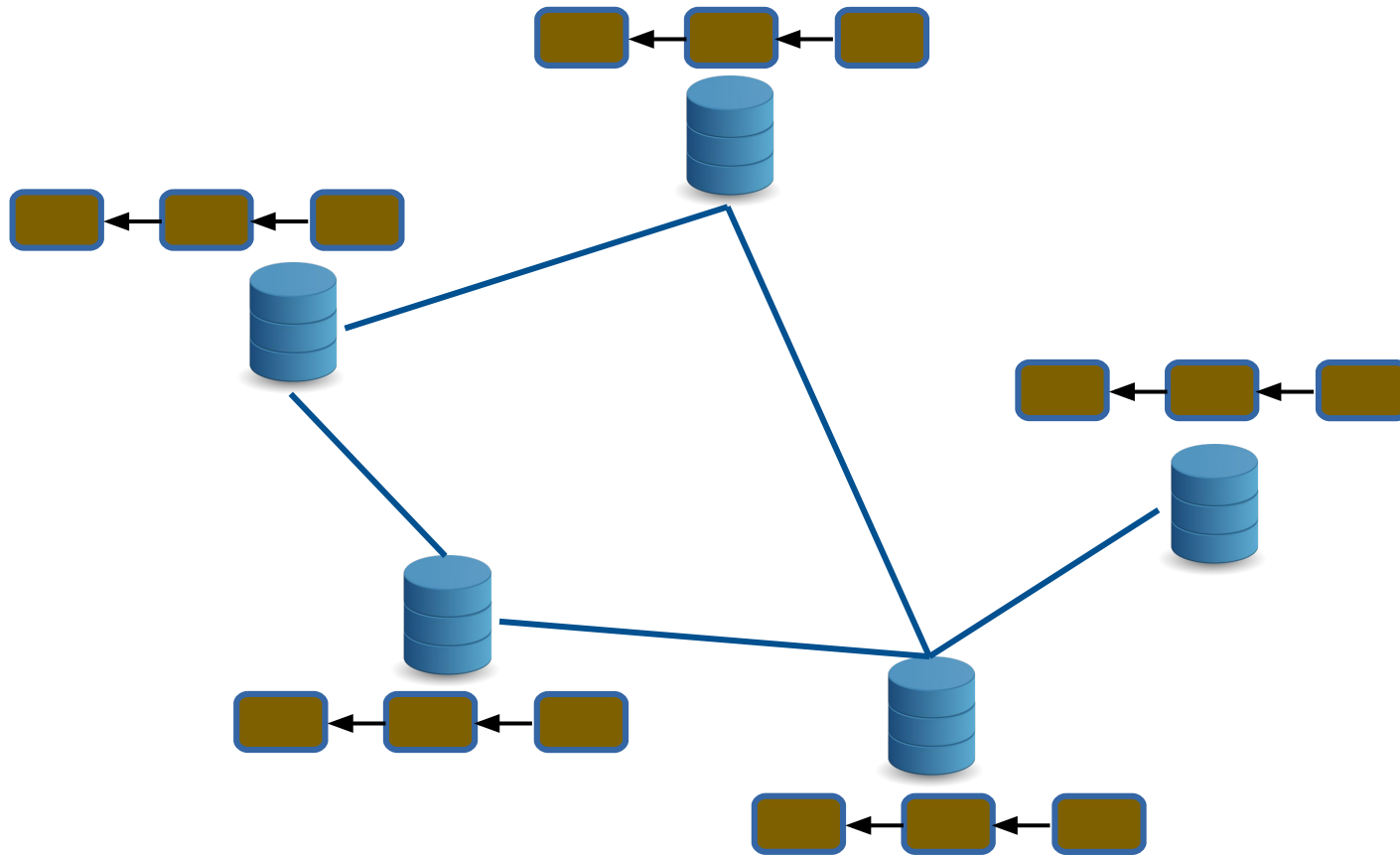
The Public Ledger



The Public Ledger

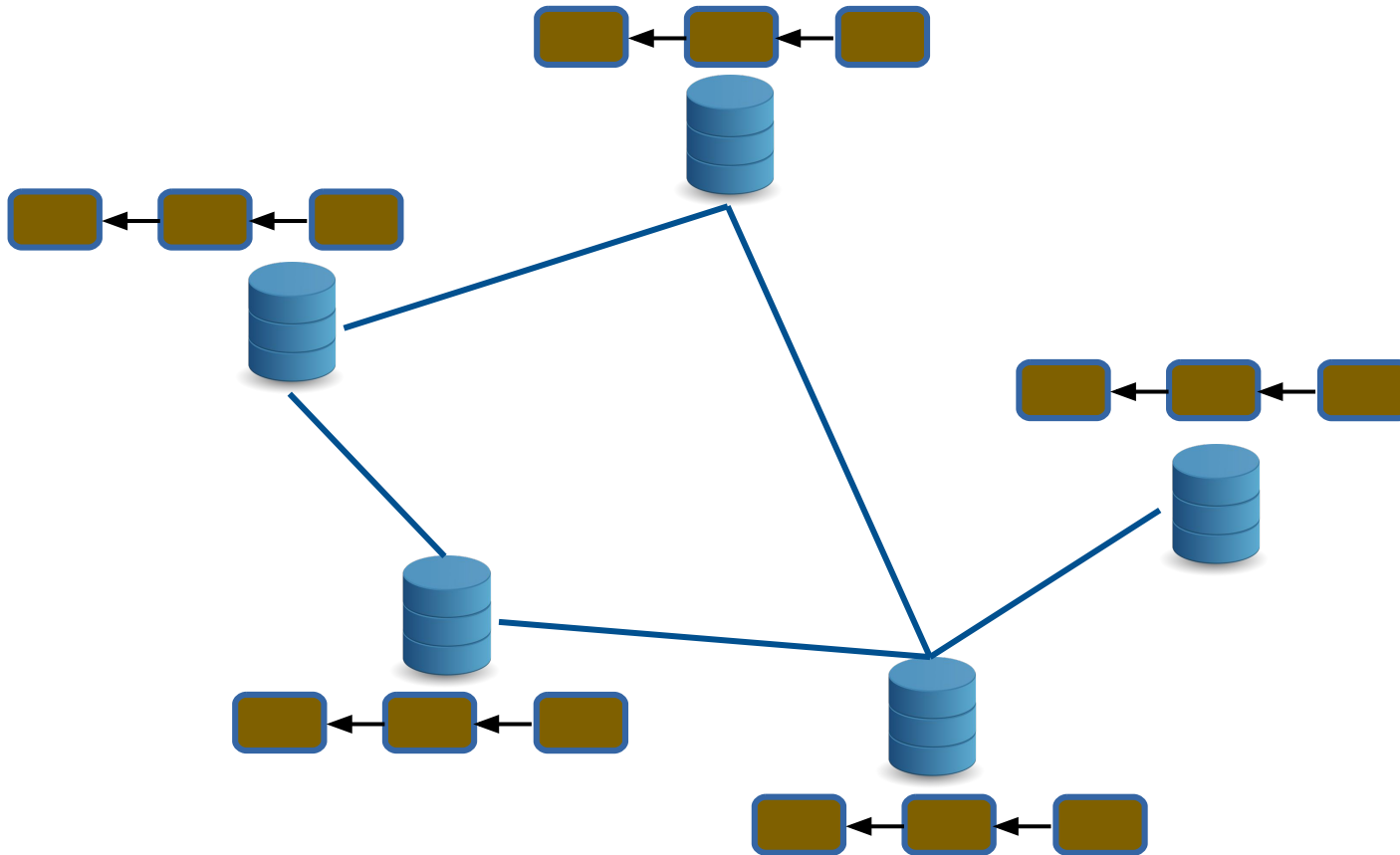


The Public Ledger

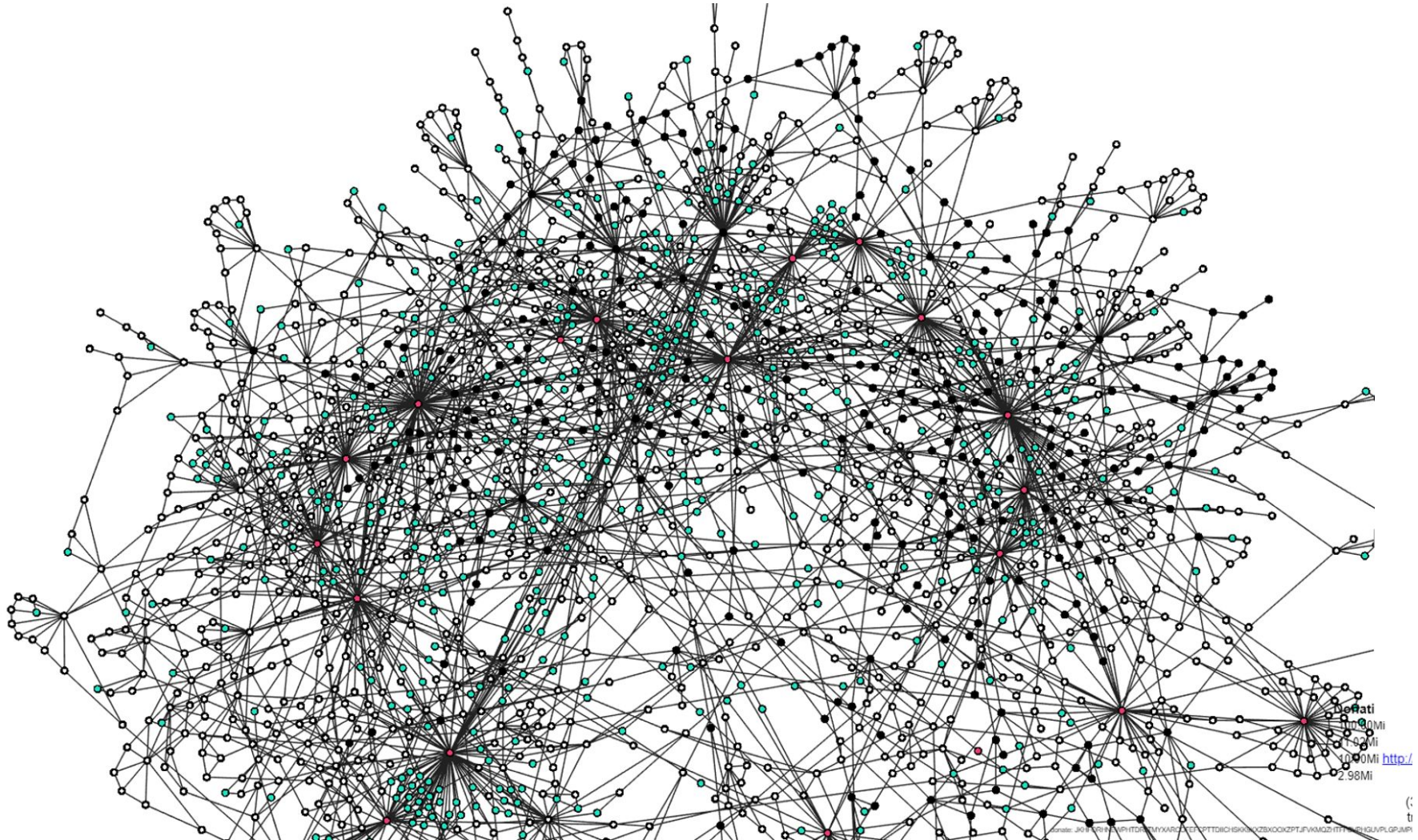


The Public Ledger

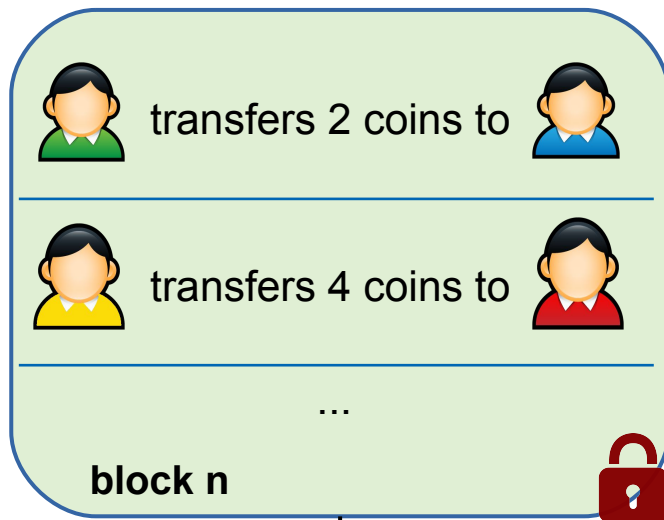
Nodes need to **agree** on the updates
(additions) to the ledger →
Consensus



The Public Ledger

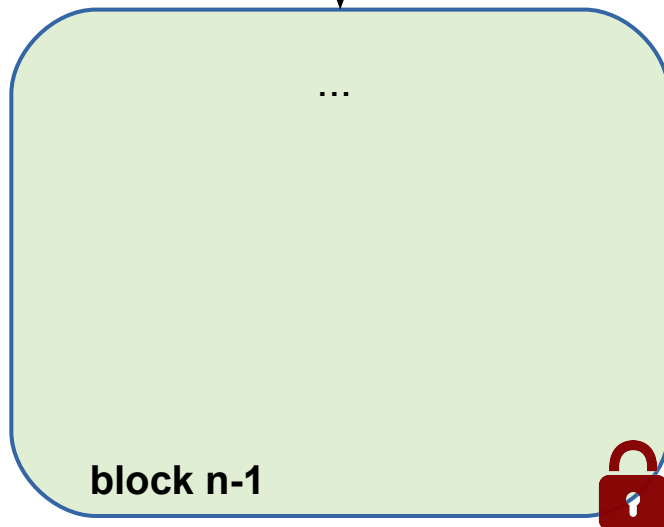


The Code

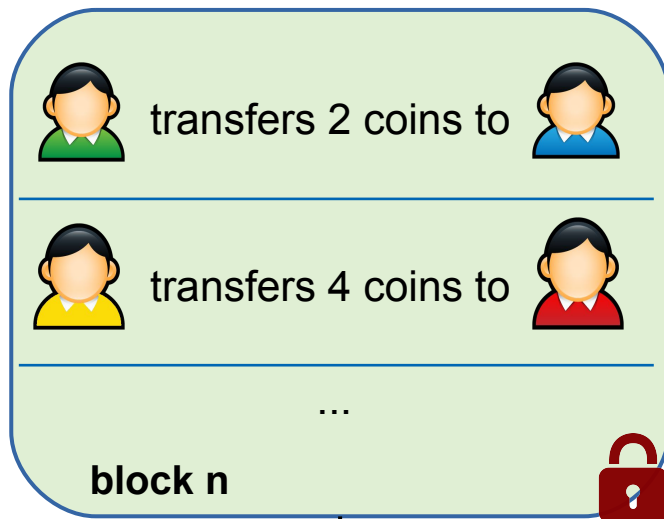


```
transaction t : {  
  a = address of   
  b = address of   
  amountToTransfer = 2;  
  transfer(a, b, amountToTransfer);  
}
```

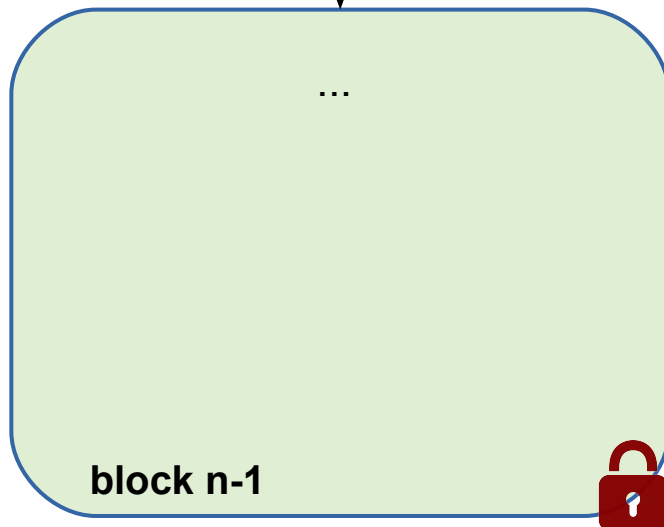
Programming code



The Code



```
transaction t : {  
  a = address of   
  b = address of   
  amountToTransfer = 2;  
  transfer(a, b, amountToTransfer);  
}
```



Programming code

Let's put on the blockchain
more complex programs

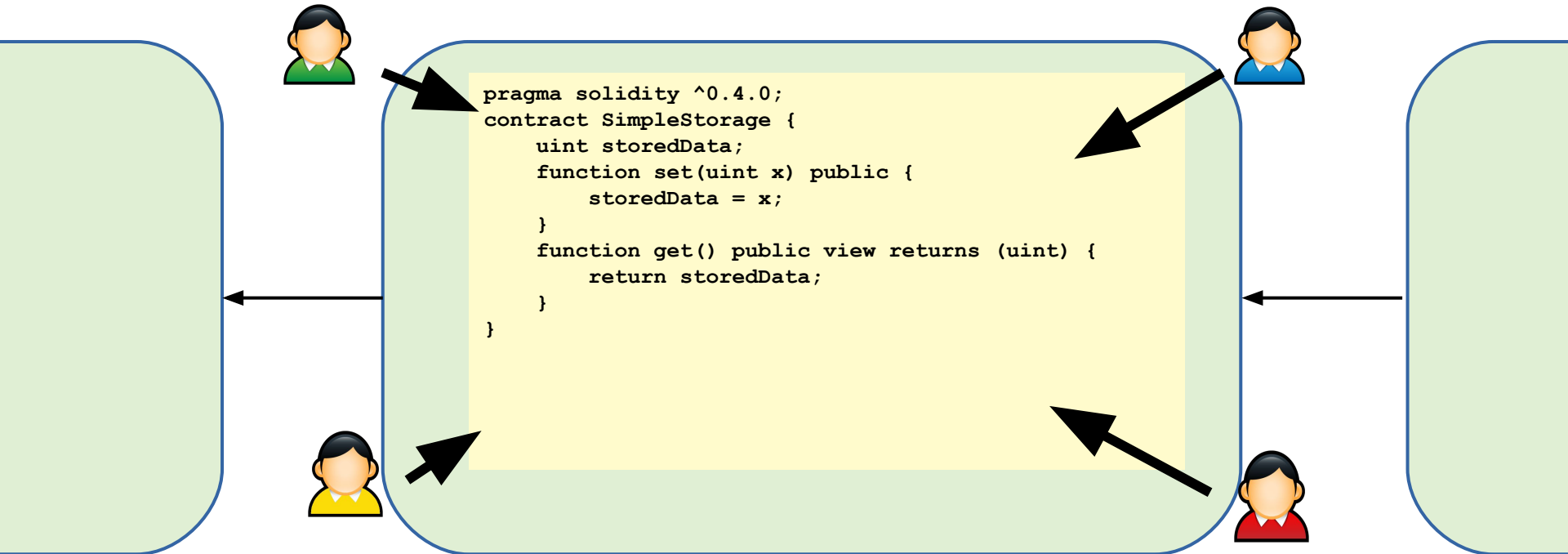


The Code → Smart Contracts



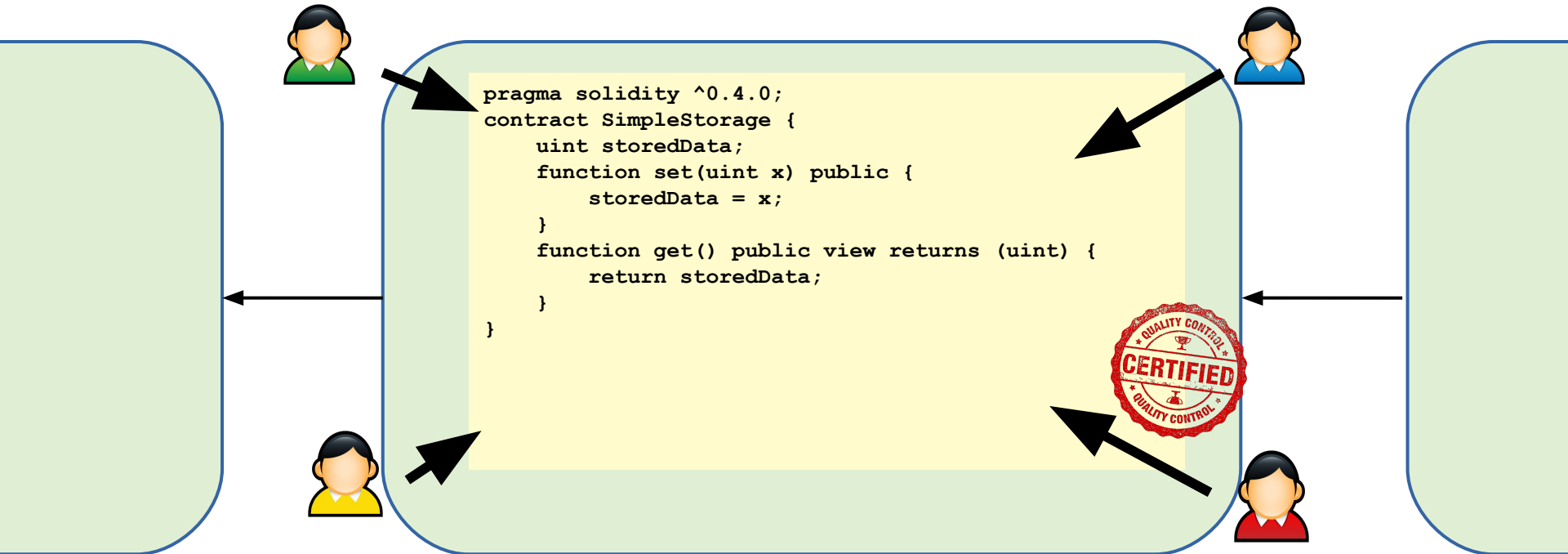
Blockchain as a platform that offers
certified services to applications → **smart contracts**

The Code → Smart Contracts



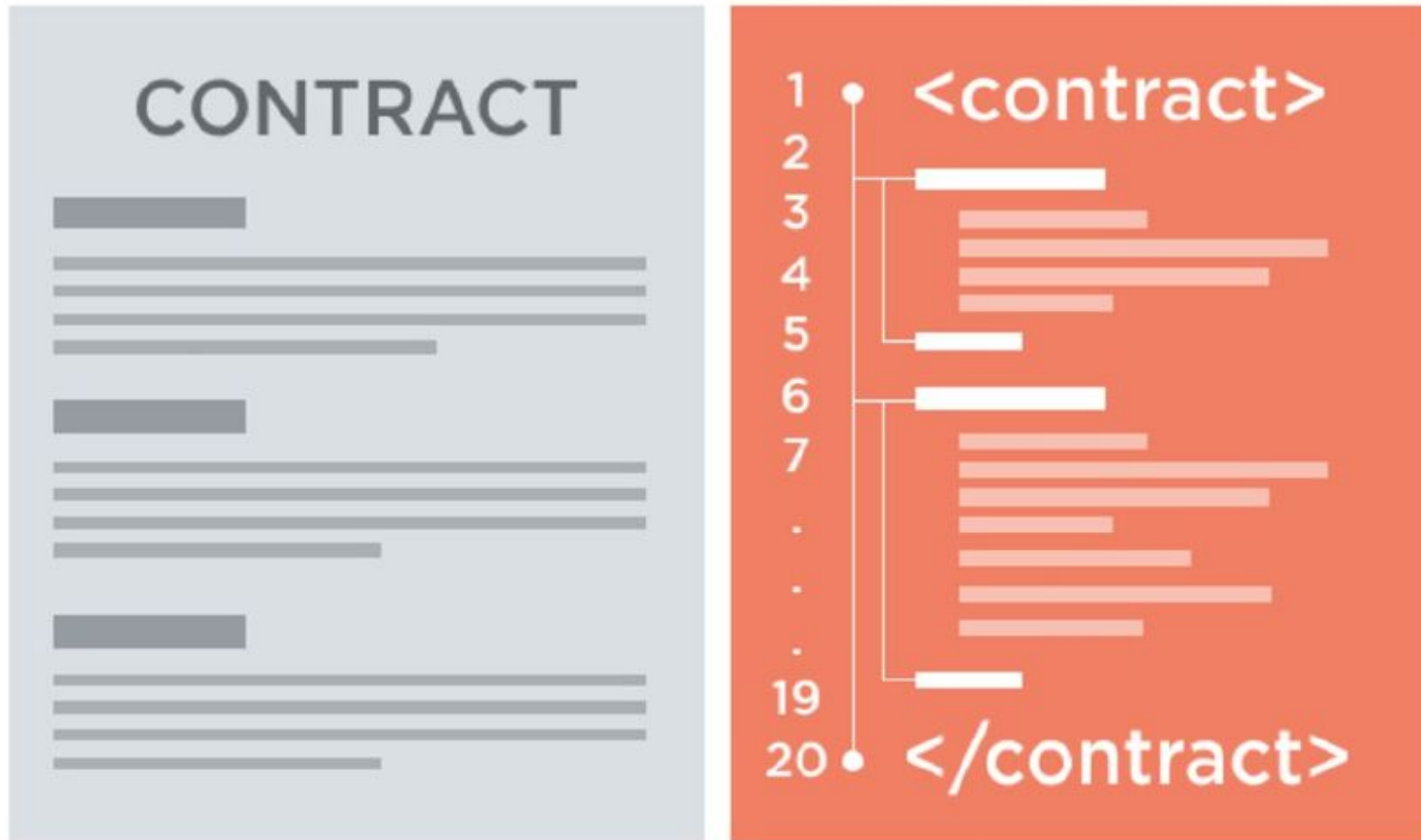
- Agreement on the execution of programs living in the blockchain
 - Cannot be modified
 - Automatically executed

The Code → Smart Contracts



- Agreement on the execution of programs living in the blockchain
 - Cannot be modified
 - Automatically executed

Legalese vs Computer Code



“The code as the law itself: self-contained, self-performed and self-enforced”
(Philip Boucher)



How Many Blockchains?

How Many Blockchains?

Public

Private

Access to the ledger

Permissionless

Permissioned

Right to modify the ledger

(Right to accept updates to the ledger)



How Many Blockchains?

Access to the ledger

Public

Private

Permissionless



ethereum

Rare

Permissioned



STELLAR



HYPERLEDGER
FABRIC

corda



energy web
CHAIN



Quorum

Right to modify the
ledger

How Many Blockchains?

Smart Contract based	No Smart Contracts
Ethereum	Bitcoin
Cardano	Litecoin
Solana	Ripple
Polkadot	
Avalanche	
IOTA	
Hyperledger (Fabric, Besu)	
Algorand	
IOTA	
Sui	



Distributed Ledger Technologies

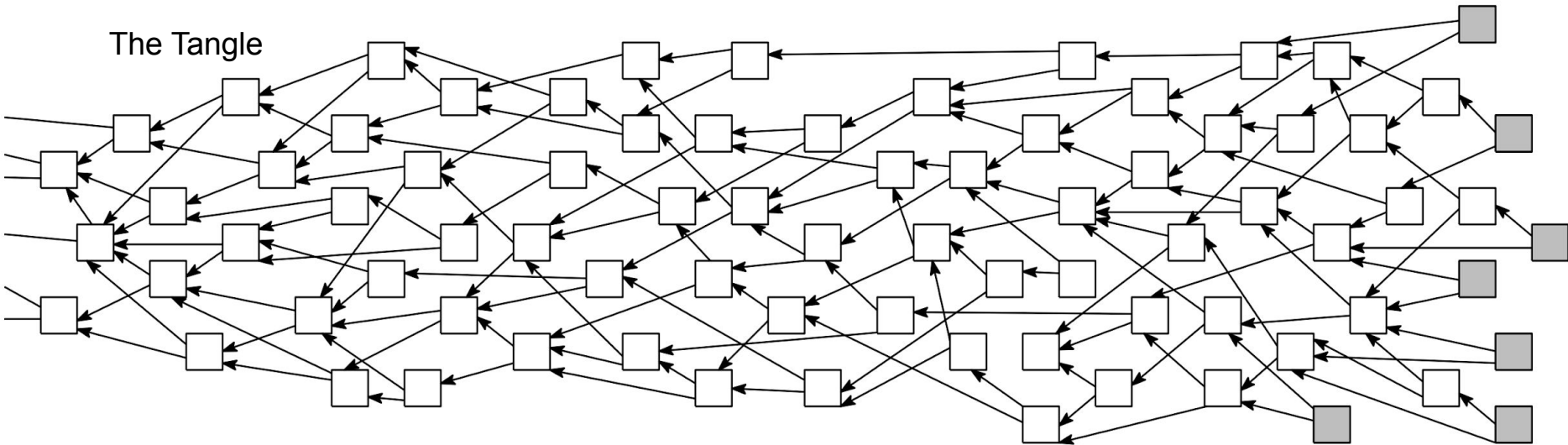
- Not all blockchains are “chains of blocks”
 - Blockchain → Distributed Ledger Technology (DLT)



An Example: IOTA

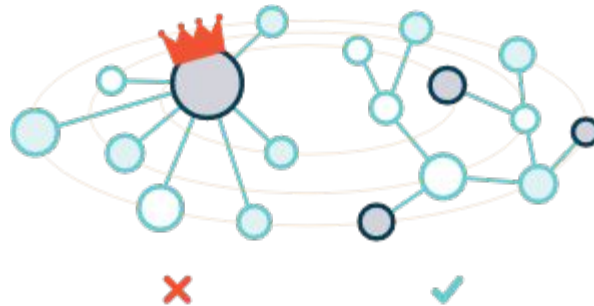
- IOTA
 - Scalability
 - Low resource requirements
 - Zero-fee transactions
 - No smart contracts

The Tangle



Not Only DLTs

- DLTs are now used with novel Decentralized File Systems
- An example - **IPFS**
 - InterPlanetary File System
 - Protocol to store hypermedia in a distributed P2P file system
 - Content-addressing to uniquely identify each file in a global namespace

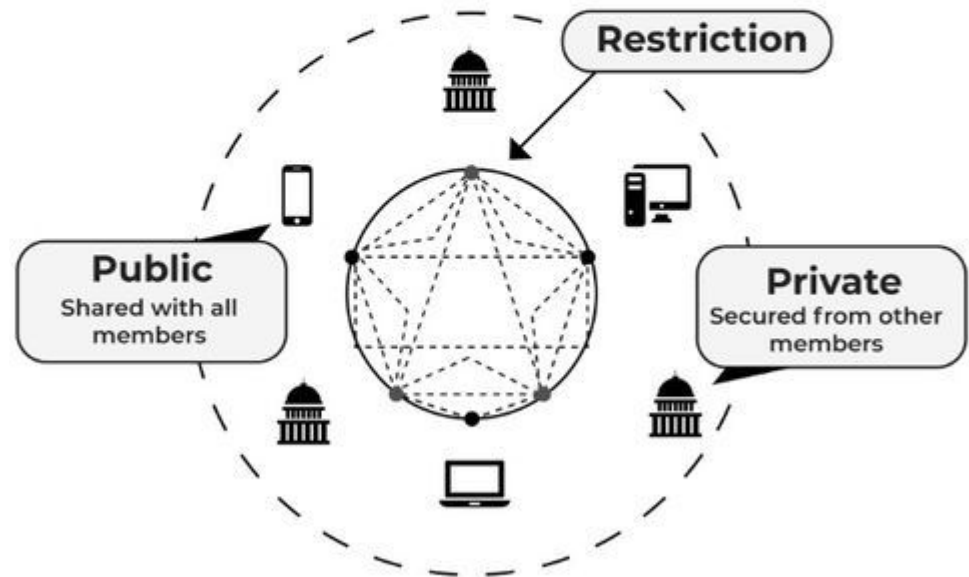


Hybrid Blockchain

In a nutshell ...

- public + private blockchains
- proofs go public
- data go private

Ex: IBM Food Trust



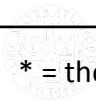
Execution Costs

Platform	Transaction fees (USD/tx)
Bitcoin	1—2
Ethereum	1.5—10
Cardano	< 0.25
Tezos	< 0.01
Hedera	0.0001
Algorand	0.0003
Solana	0.0002
Avalanche	0.03



Performance

Platform	Throughput (tx/s)	Latency
Bitcoin	3-7	1h
Ethereum	10-100	>13s
Cardano	17-20	2m
Tezos	180 [*]	30h
Hedera	400,000 [*]	10-20s
Algorand	6,000	4s
Solana	4000	1-3
Avalanche	260-7000	<1s

 ALMA MATER STUDIO FORUM
* = theoretical estimate

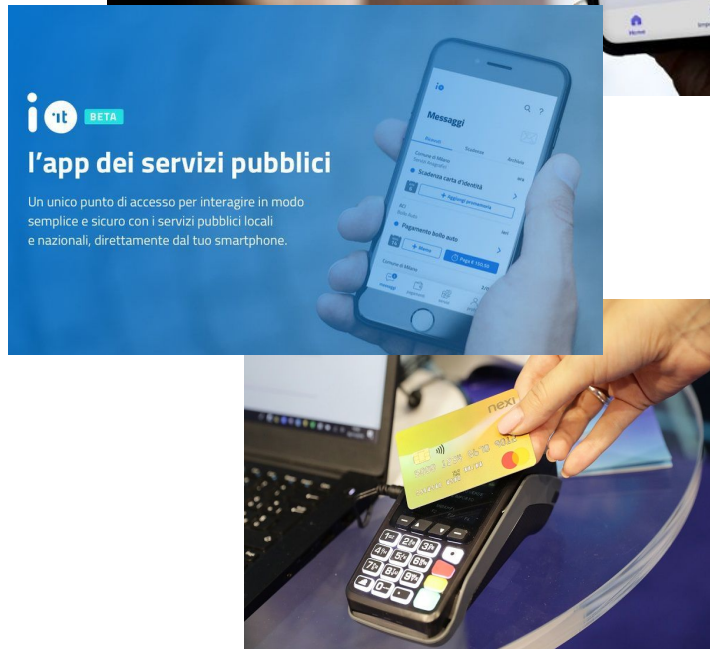
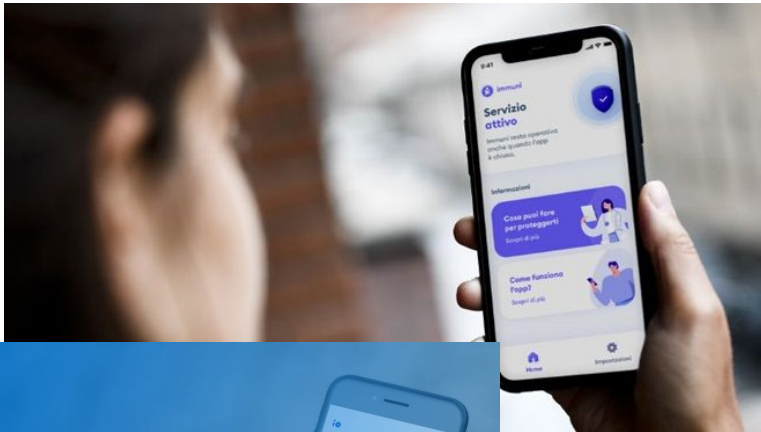
Environmental Impact

Platform	Emissions (Wh/tx)
Bitcoin	$360 \times 10^3 - 3700 \times 10^3$
Ethereum	18 — 557
Cardano	12—378
Tezos	0.36—11
Hedera	0.02—0.04
Algorand	0.17 — 5.34
Solana	0.166
Avalanche	4.76

Sources: Moritz Platt et al., “The Energy Footprint of Blockchain Consensus Mechanisms Beyond Proof of Work,” 2021; Ulrich Gellersdörfer et al., “Energy Efficiency and Carbon Footprint of PoS Blockchain Protocols,” and “The Merge: Implications on the Electricity Consumption and Carbon Footprint of the Ethereum Network,” 2022.

Some use cases

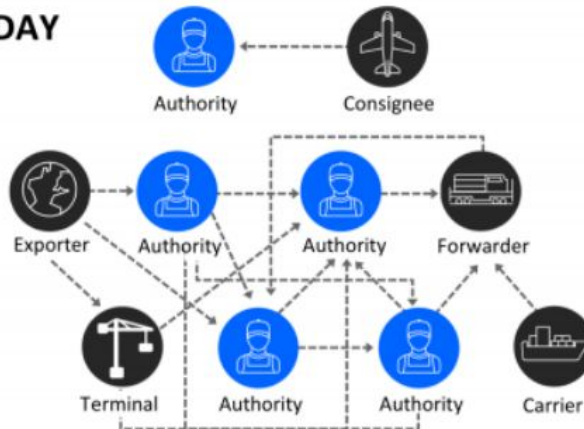
Traceability



Digitizing Global Trade with Maersk and IBM



TODAY



- Inconsistent information across organizational boundaries and “blind spots” throughout the supply chain hinder the efficient flow of goods
- Complex, cumbersome, and costly peer-to-peer messaging
- Manual, time-consuming, paper-based processes
- Risk assessments often lack sufficient information; clearance processes subject to fraud
- The administrative cost of handling a container shipment is comparable to the cost of the actual physical transport

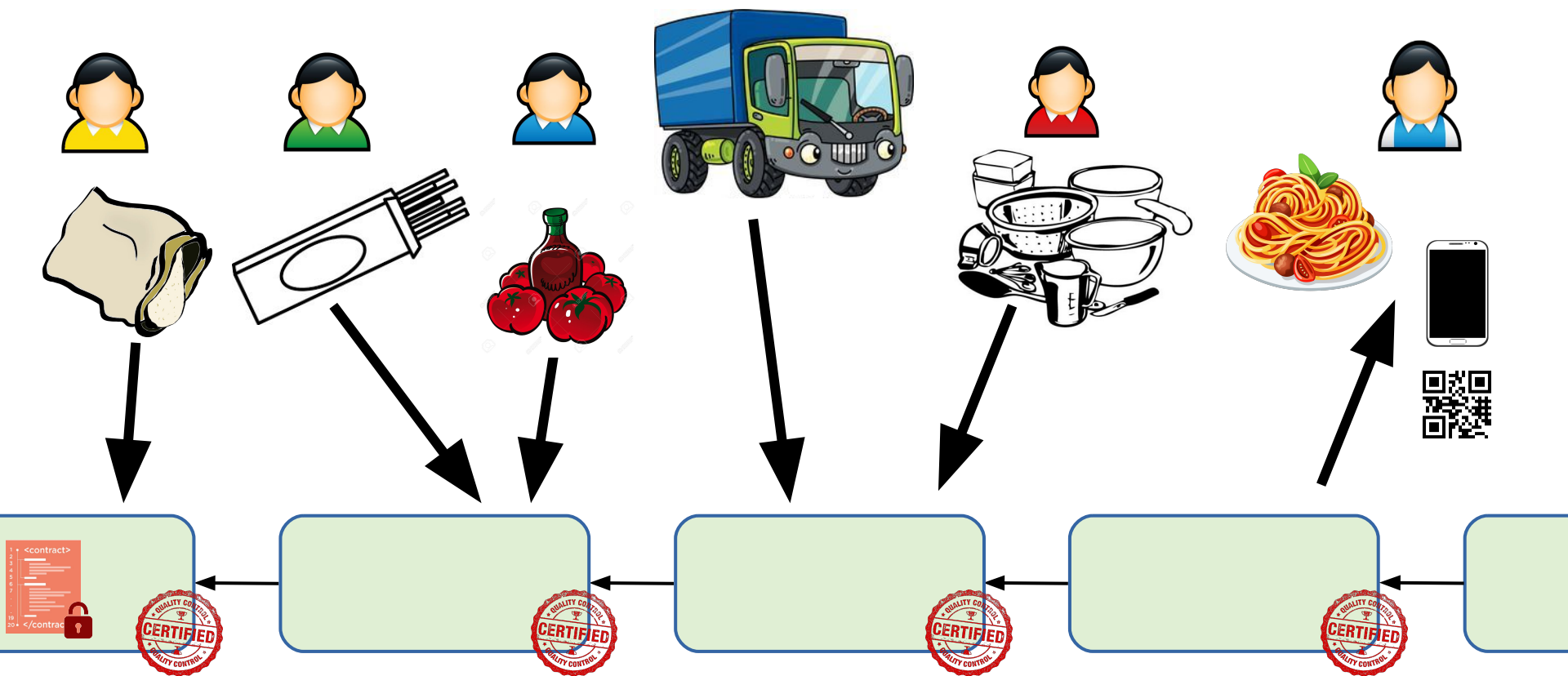
FUTURE



- Fast, secure access to end-to-end supply chain information; single source of the truth
- Verifiable authenticity and immutability of digital documents
- Trusted cross-organizational workflows
- Better risk assessments and fewer unnecessary interventions
- Far lower administrative expenses and elimination of costs to move physical paper across international borders



Traceability in Agri-Food



Some prototypes

- AgroChain: Agricultural Supply Chain Dapp With Micro-Finance
<https://github.com/Kerala-Blockchain-Academy/AgroChain>
- Hackinators-Farming-Dapp
(very basic ...)
<https://github.com/ShubhamKarala/Hackinators-Farming-Dapp>



Examples in Italy



barbara carfagna @barbarcarfagna - 27 gm
Regala la prima Falanghina al mondo certificata e tracciata attraverso Blockchain
Ethereum

09 GENNAIO 2019

La filiera qualità di Carrefour accelera con la blockchain

GDO

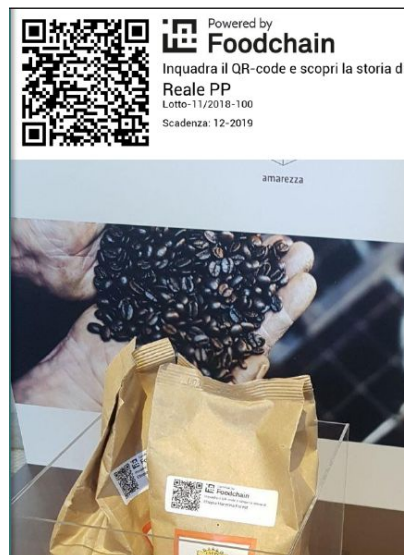
Entro l'anno verrà usata per agrumi, vongole, pomodori e lattiero caseario

Enrico Netti

Carrefour Italia accelera nell'uso della

getto è partito da poco e per le vendite del pollo, prodotto al centro della prima applicazione, abbiamo registrato il raddoppio dei ricavi oltre ad un aumento delle visite alla pagina accessibile con Qr code» sottolinea Panzeri. Come interpretare la corsa verso la blockchain della Gdo e dei produttori di food? «Tutte le filiere complesse con diversi attori che giocano un ruolo preciso nella realizzazione del pro-

una riduzione del 15% dei tempi di controllo qualità, il miglioramento dei margini del prodotto del 5%, la certificazione dell'impronta di sostenibilità, la riduzione dei tempi complessivi di processo del 20% grazie alla digitalizzazione di filiera». Oltre a questi plus le insegne, tra cui Carrefour Italia che usa la EY Op-sChain food traceability basata sulla piattaforma pubblica di Ethereum



09 GENNAIO 2019

8/9 di 32

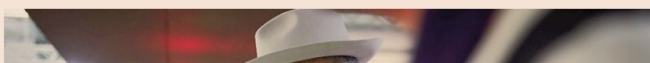
Economia & Imprese

Moda come gli alimenti, vince se è tracciabile

FIERE

Marino Vago, presidente

prossimo autunno-inverno), una "fiera" che attira circa 25mila buyer da 100 Paesi. «Faccio appello agli operatori del settore a investire in



IL COMPARTO

05.11.19

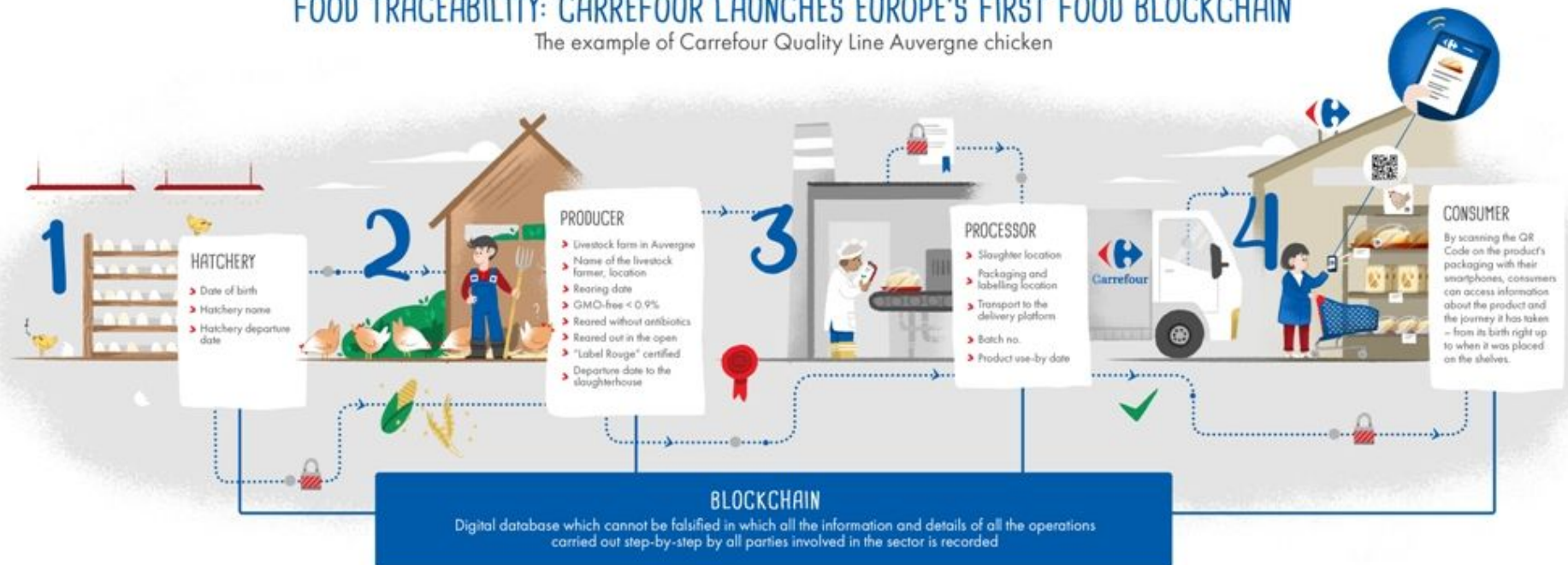
Mansi, presidente del Centro di Firenze per la moda italiana (Cfmi) che controlla la società operativa Pitti Immagine nata 20 anni fa, un



Carrefour

FOOD TRACEABILITY: CARREFOUR LAUNCHES EUROPE'S FIRST FOOD BLOCKCHAIN

The example of Carrefour Quality Line Auvergne chicken



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Carrefour

rgnefqc.carrefour.com

Informations sur votre produit



Poulet fermier d'Auvergne jaune filière qualité Carrefour

A consommer jusqu'au : **11/03/2018**

*A conserver au froid jusqu'à 4°C
A cuire à coeur*



rgnefqc.carrefour.com

Informations sur votre produit



ELEVAGE
du 11/12/2017
au 01/03/2018



Par :
GENDRE
FREDERIQUE
ANTHONY ET
JOCELYN

Lieu :
CHIRAT L EGLISE

Elevage :
Fermier, en plein
air, label rouge
certifié par
Certipaq

rgnefqc.carrefour.com

Informations sur votre produit



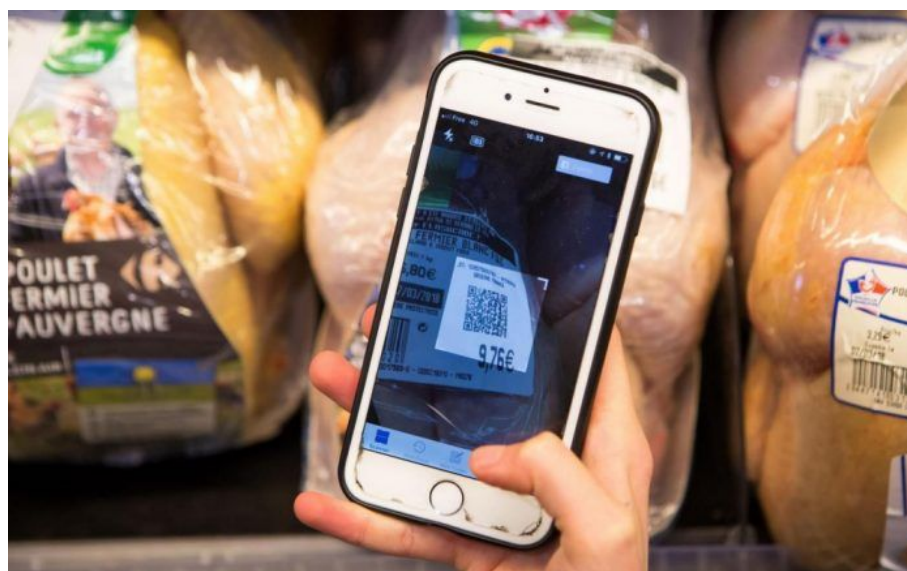
ABATTAGE
le 01/03/2018

Abattoir :
Arrivé Auvergne
dans le respect
des normes
d'hygiène et de
l'animal

Lieu :
Saint-Germain-
des-Fossés
(Allier)

Le :
01/03/2018 à
l'âge de 81 jours

[En savoir +](#)



Riseria Campanini



LEGGI IL **NUMERO DI LOTTO** SUL FONDO DELLA SCATOLA

L

21 1/070|

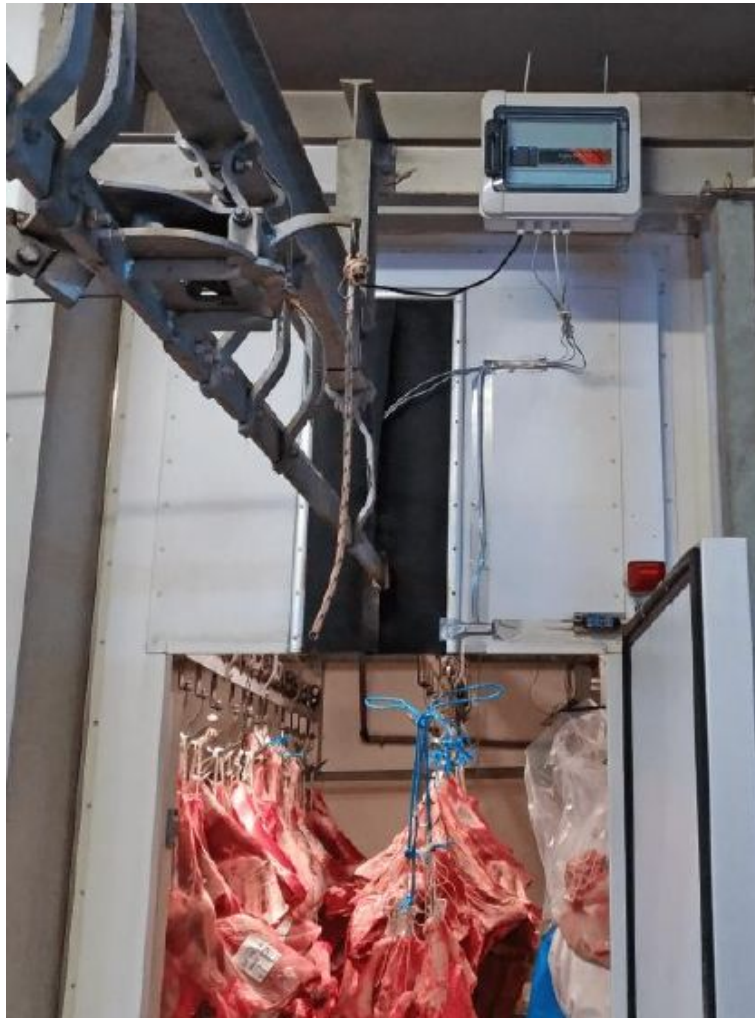
INVIA

<https://agrchain.riseriacampanini.it/bio/>



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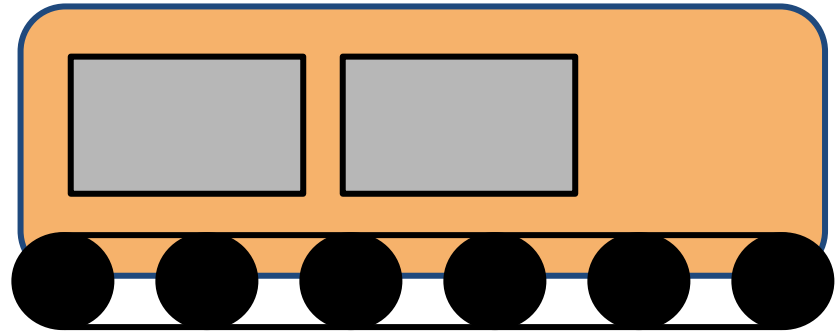
Cold Chain Maintenance



Blockchain and Supply Chain

Production company

A

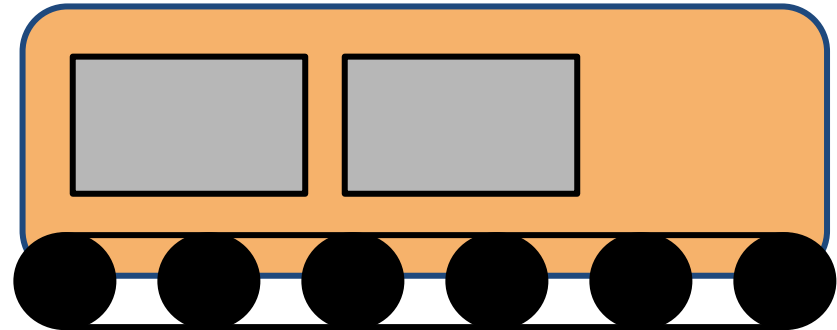


Product developed by company A



Components

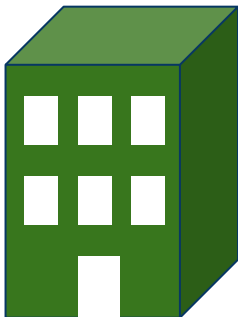
A



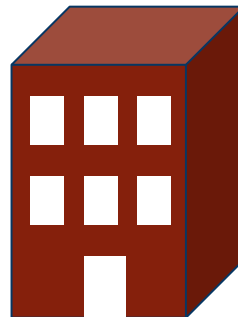
Components produced by other companies



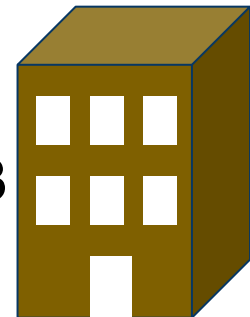
B1



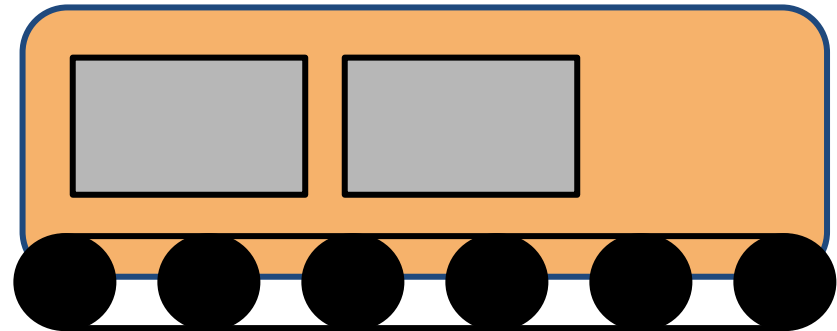
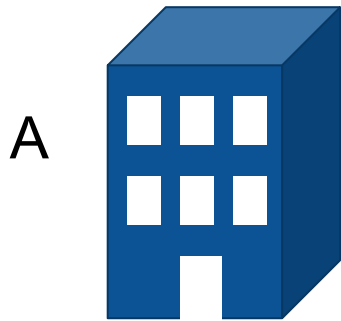
B2



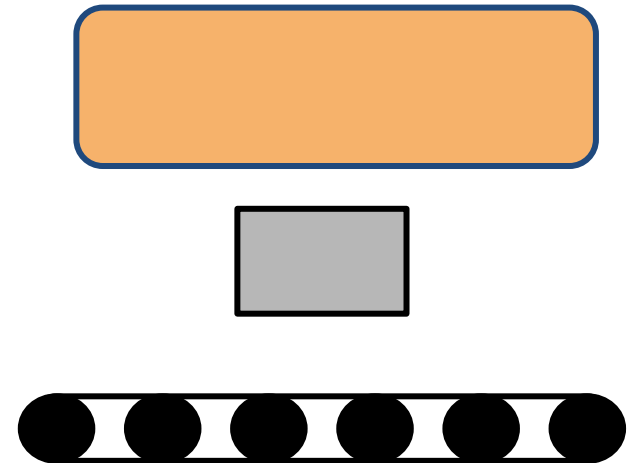
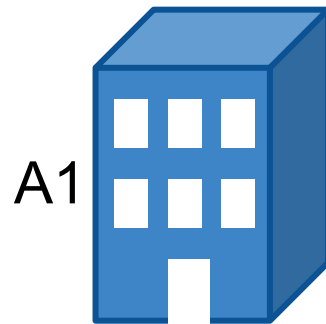
B3



Components

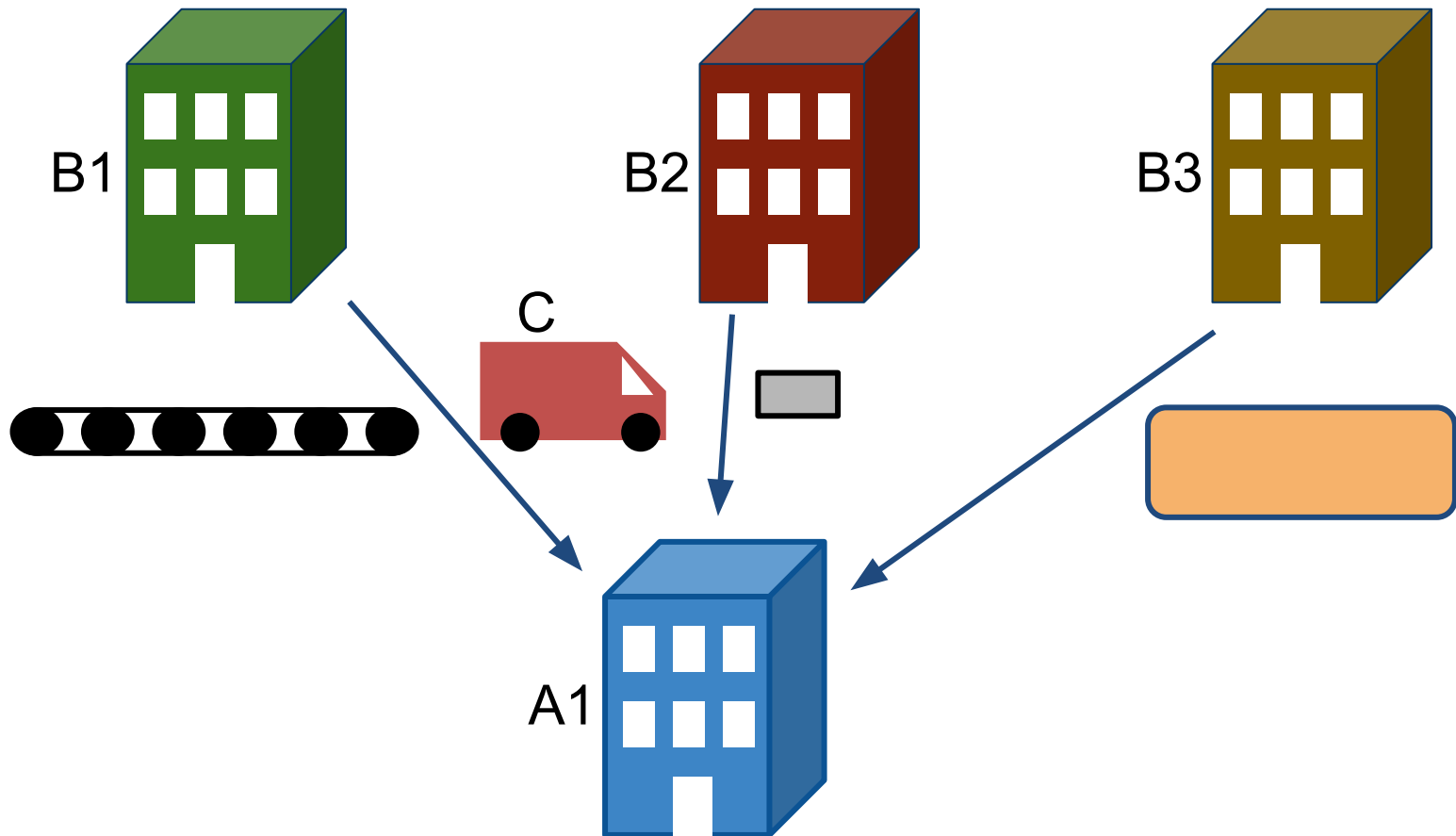


Components assembled in a specific factory (held by A)

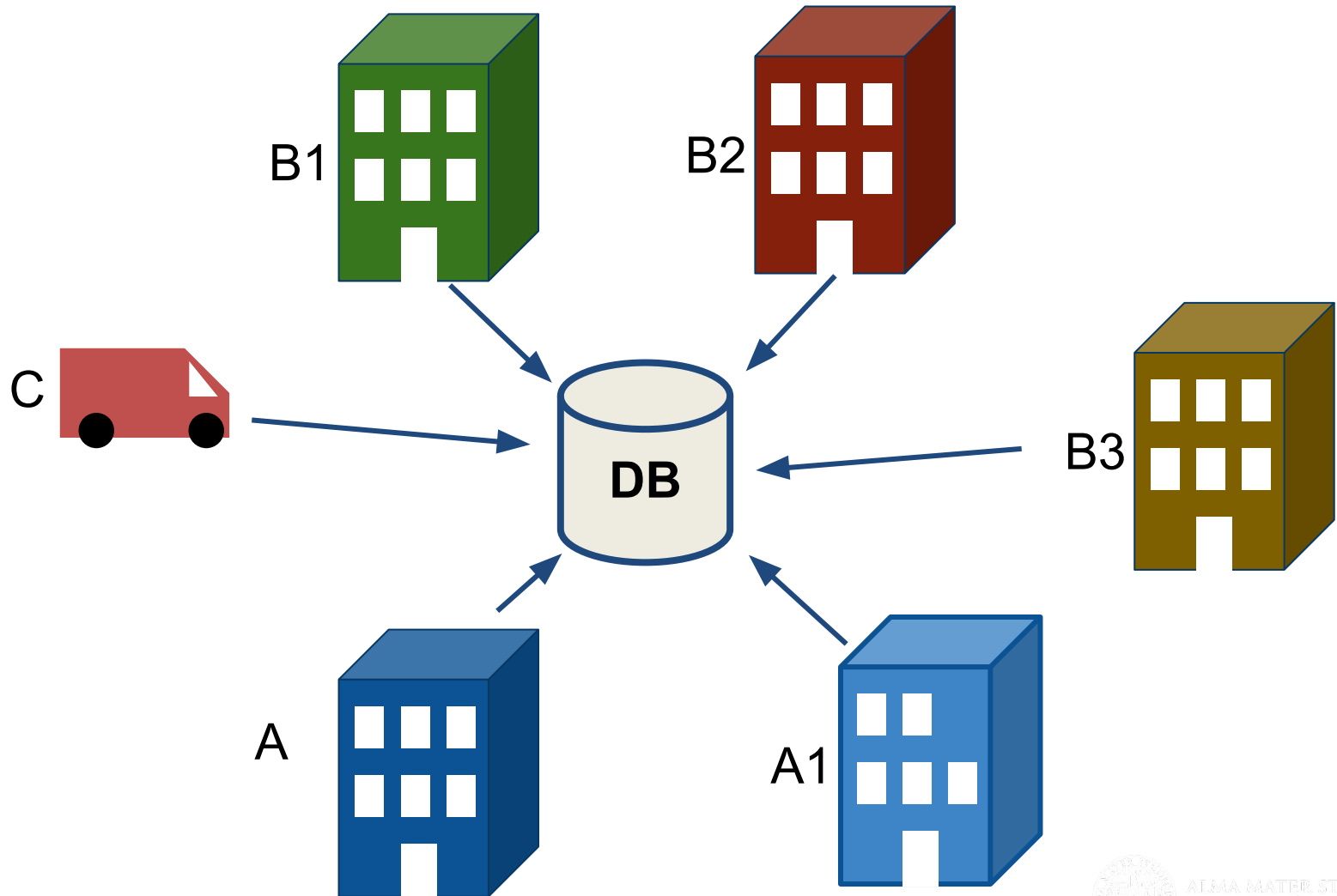


Components

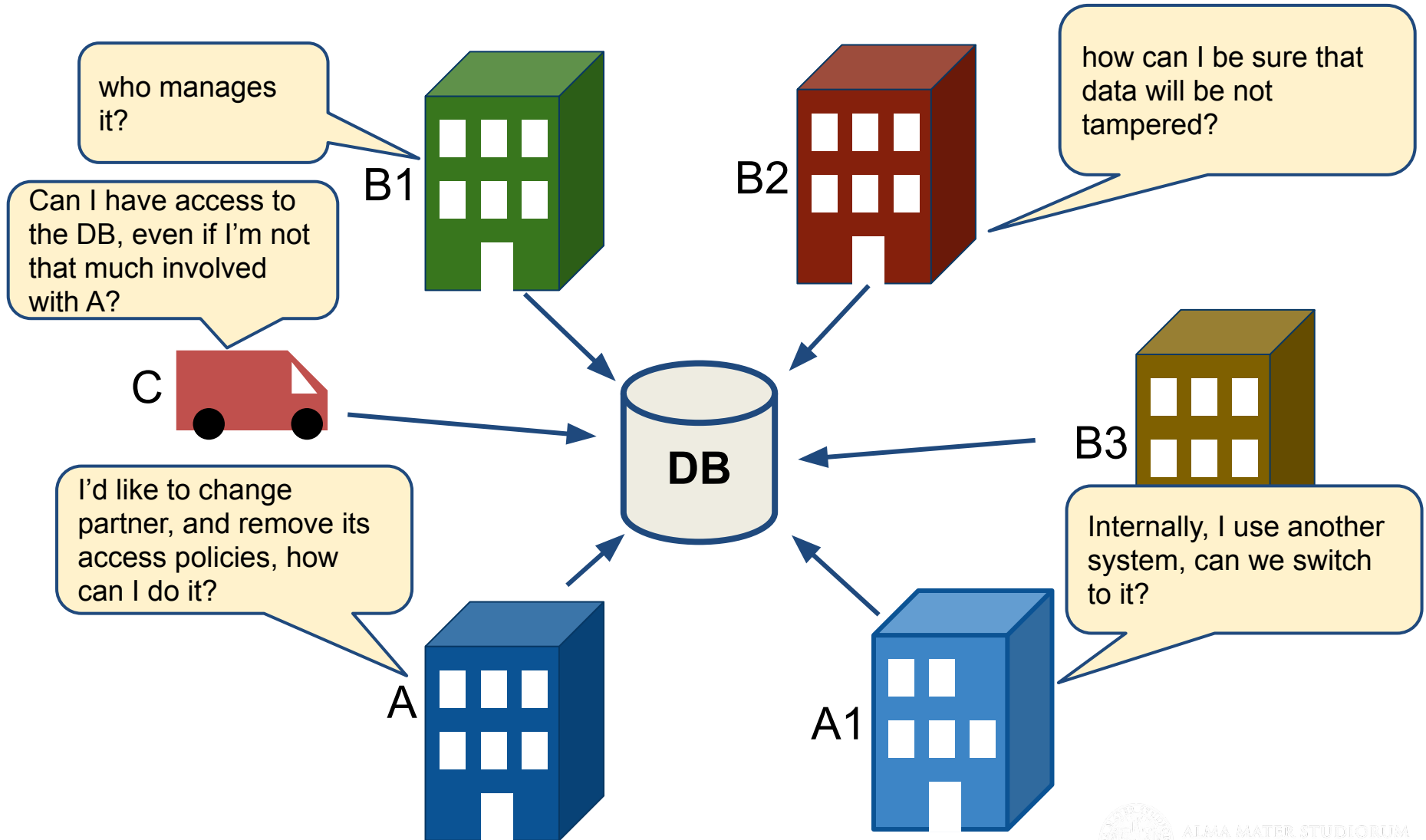
Components delivered by a carrier



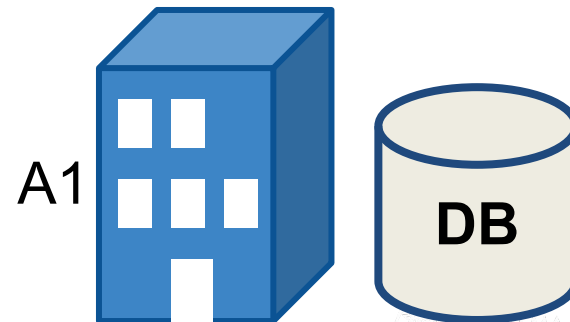
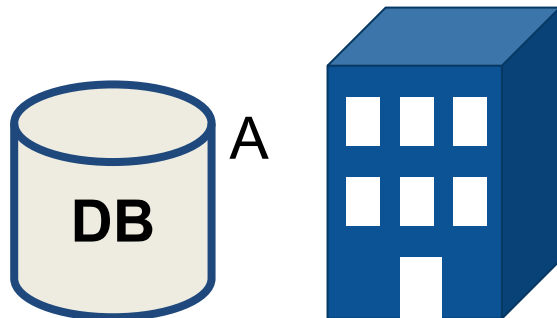
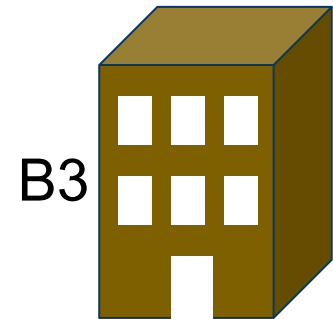
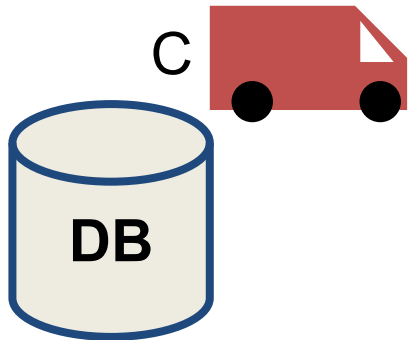
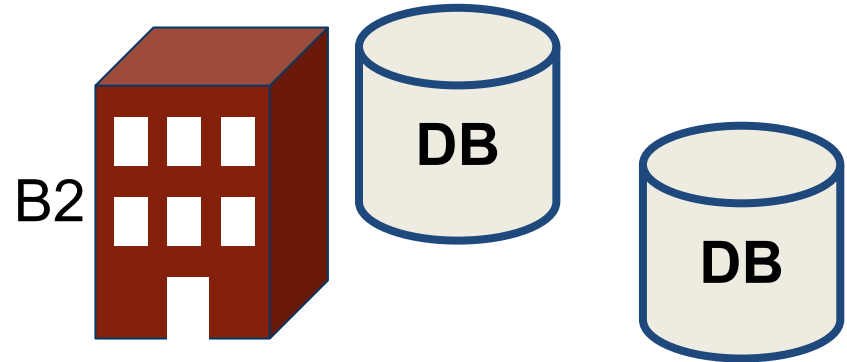
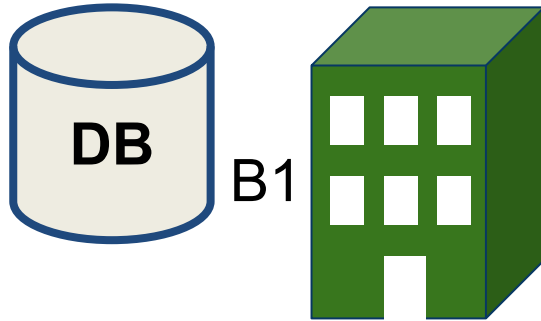
Let's Trace everything ... with a db



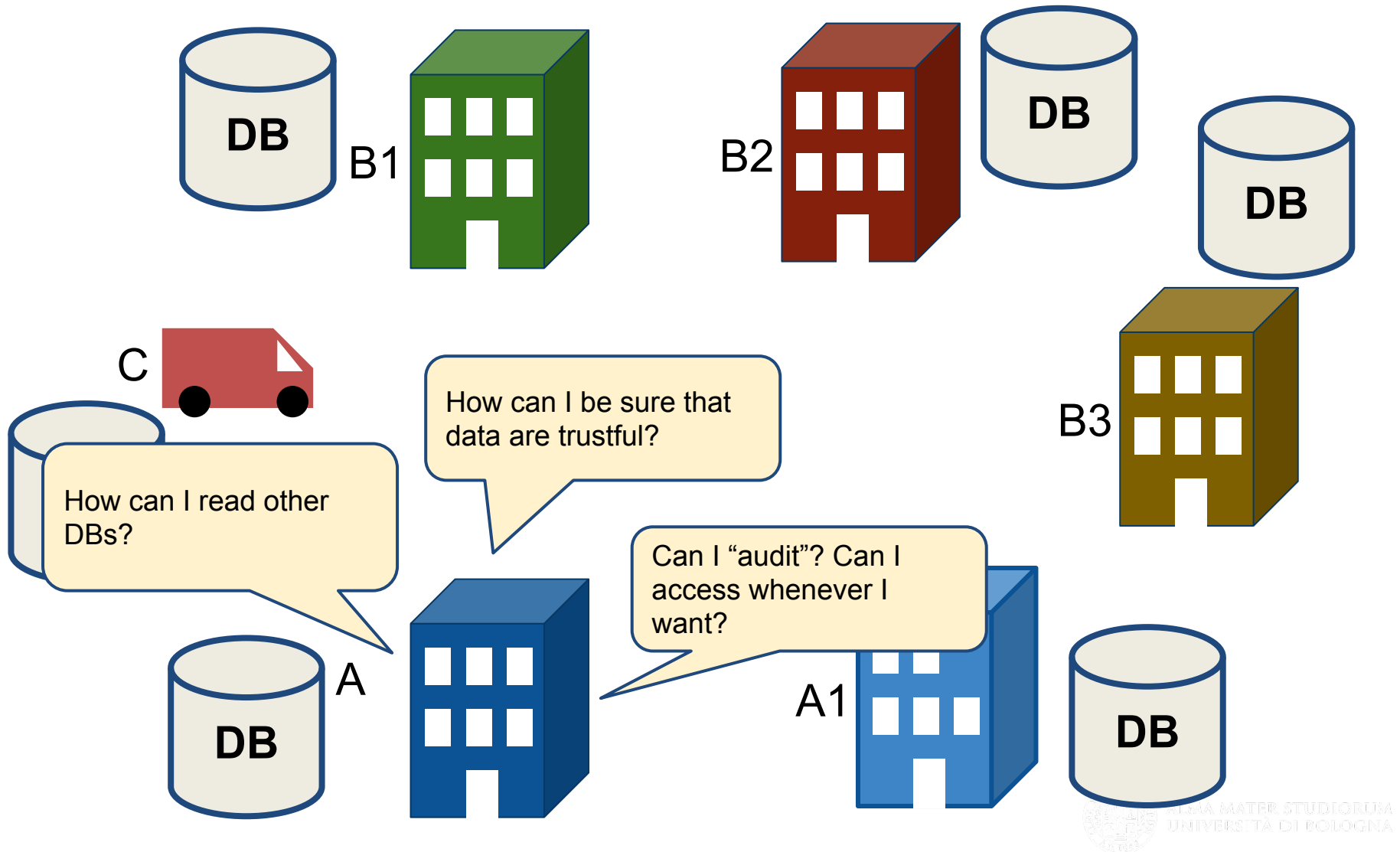
Let's Trace everything ... with a db



Let's Trace everything ... with many DBs



Let's Trace everything ... with many DBs



Let's Trace everything ... with the blockchain

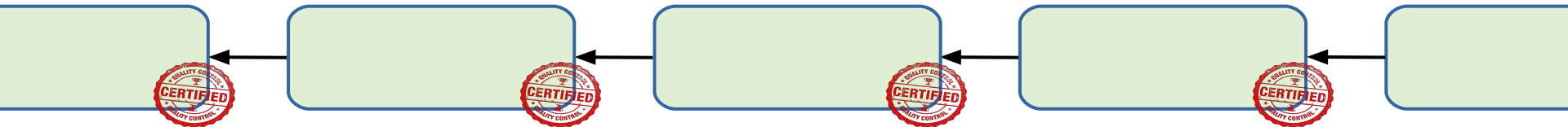
A



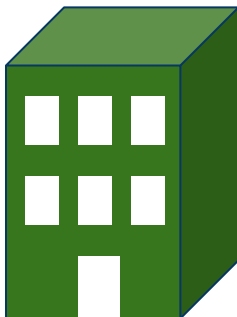
A1



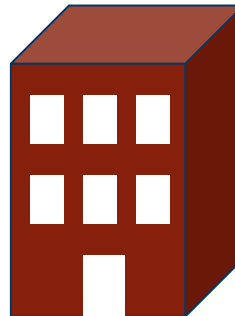
C



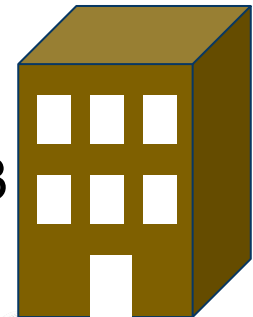
B1



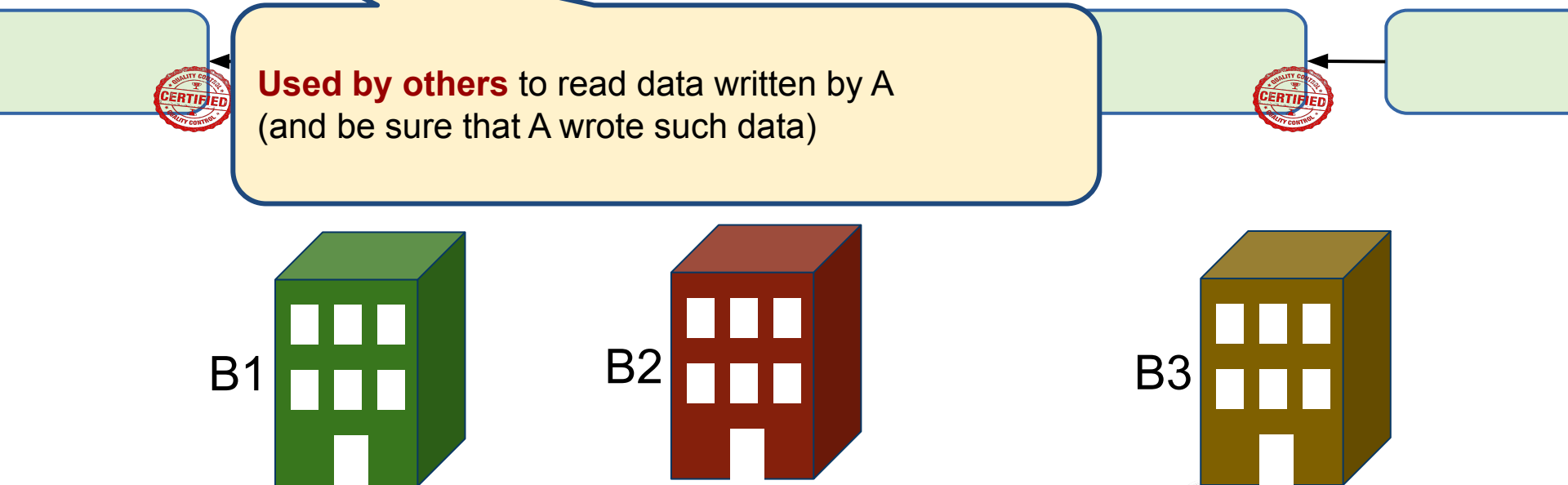
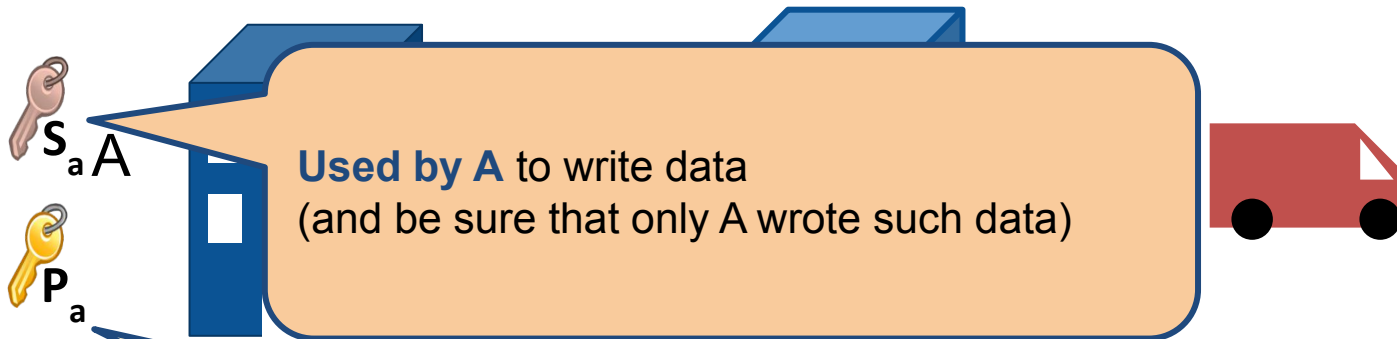
B2



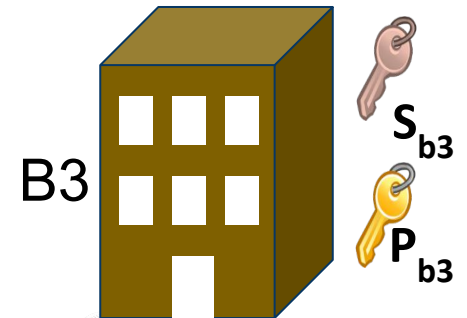
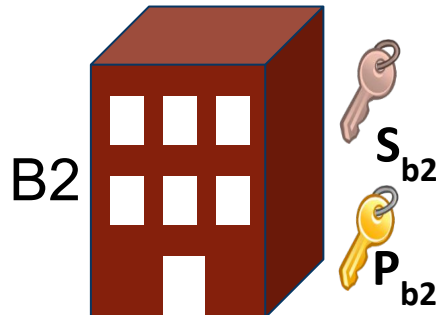
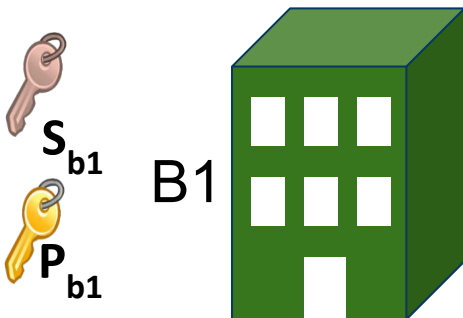
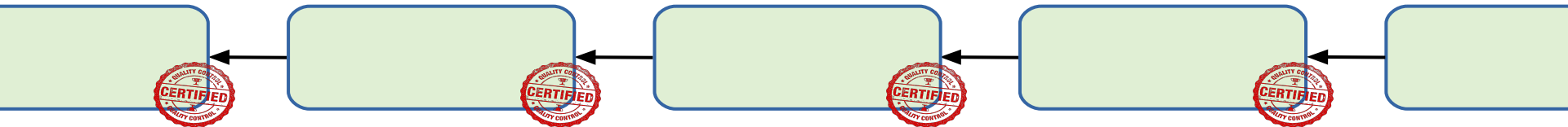
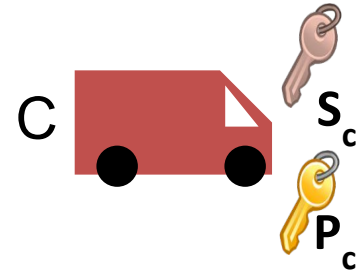
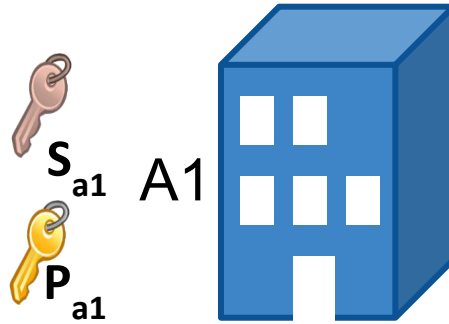
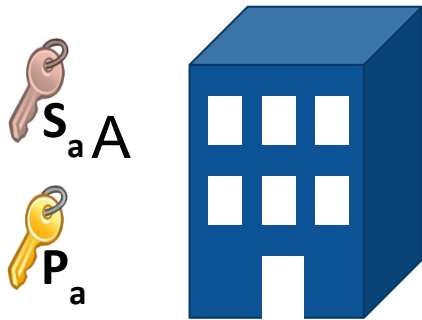
B3



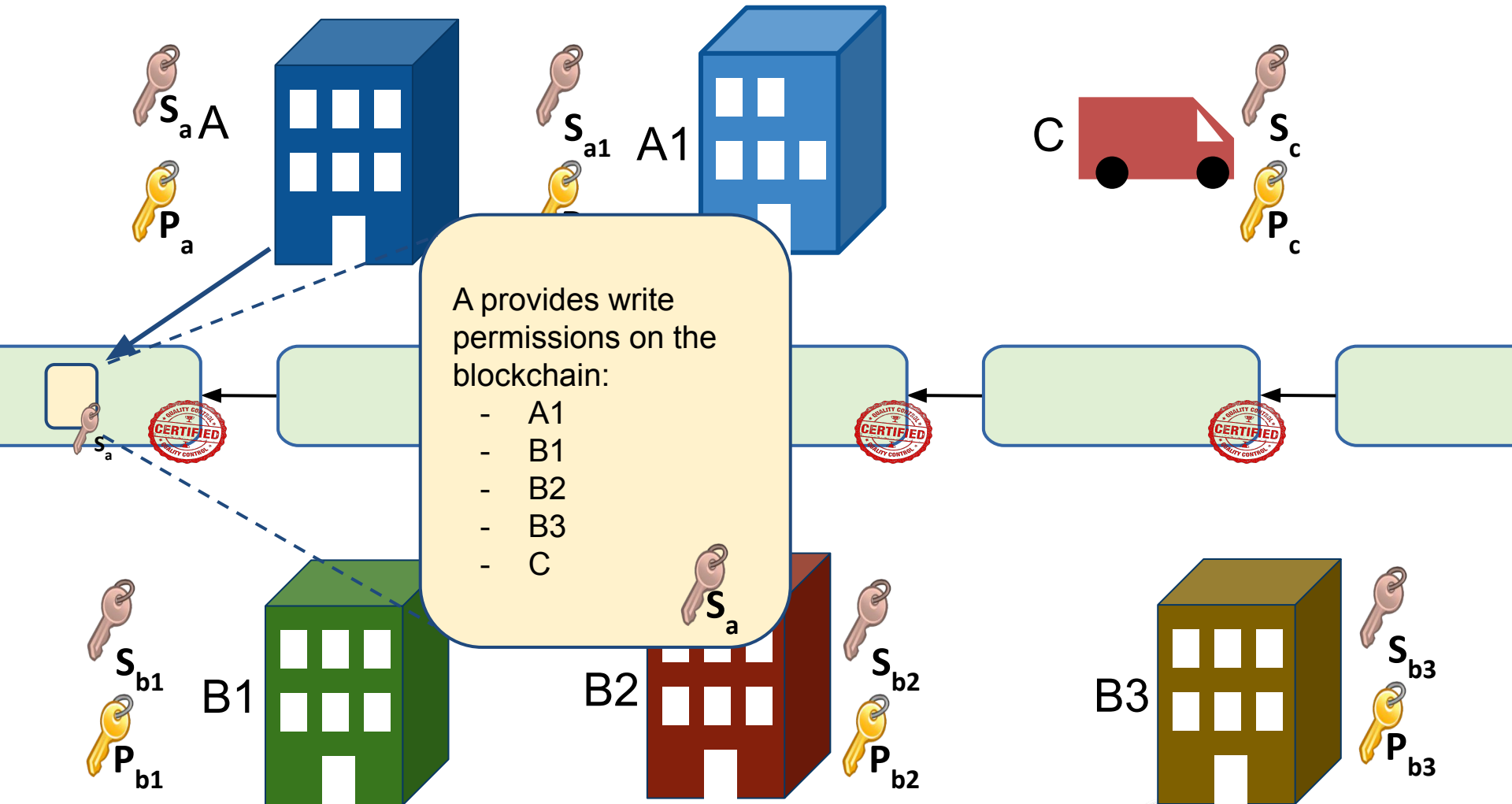
Let's Trace everything ... with the blockchain



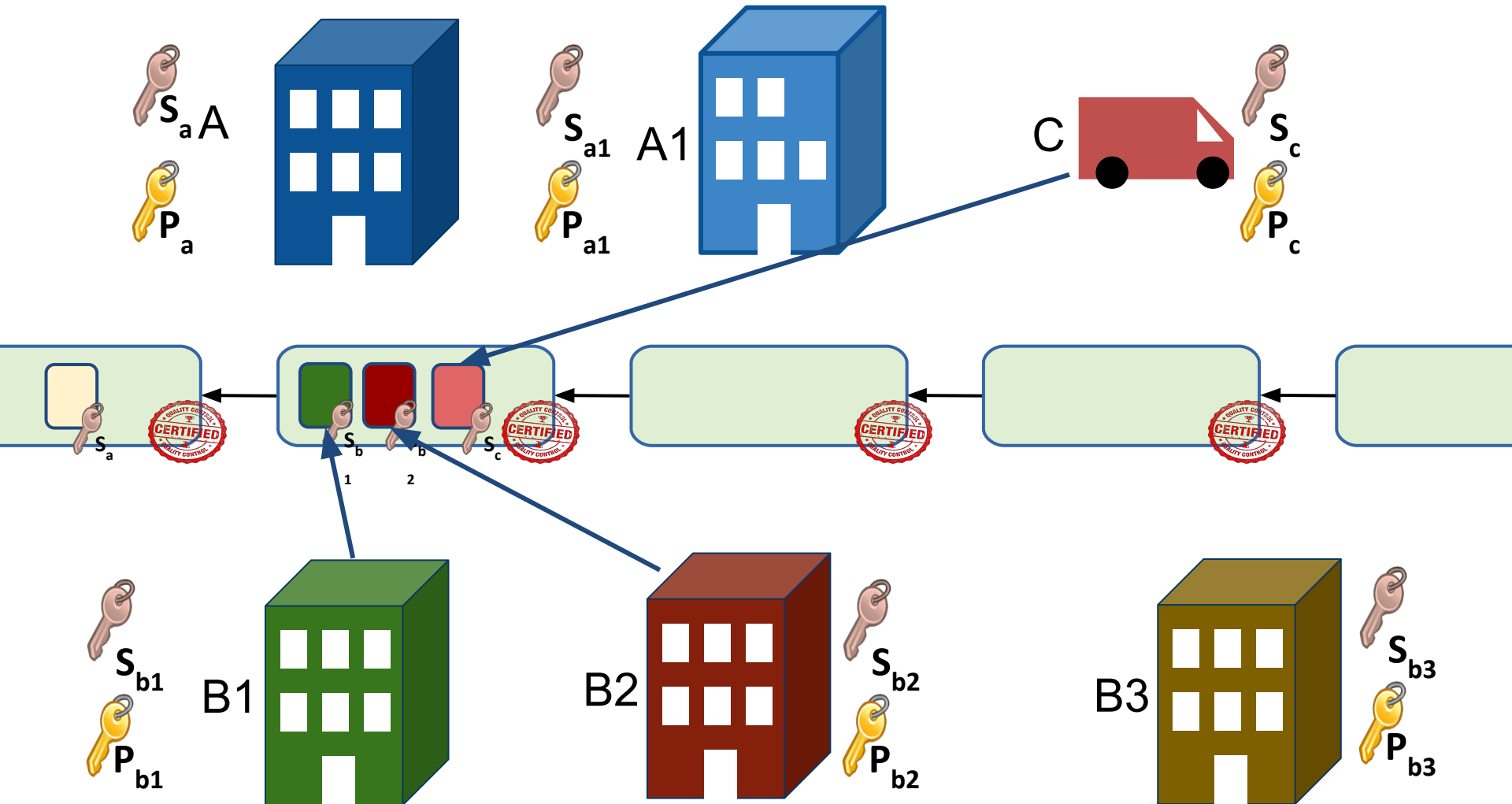
Let's Trace everything ... with the blockchain



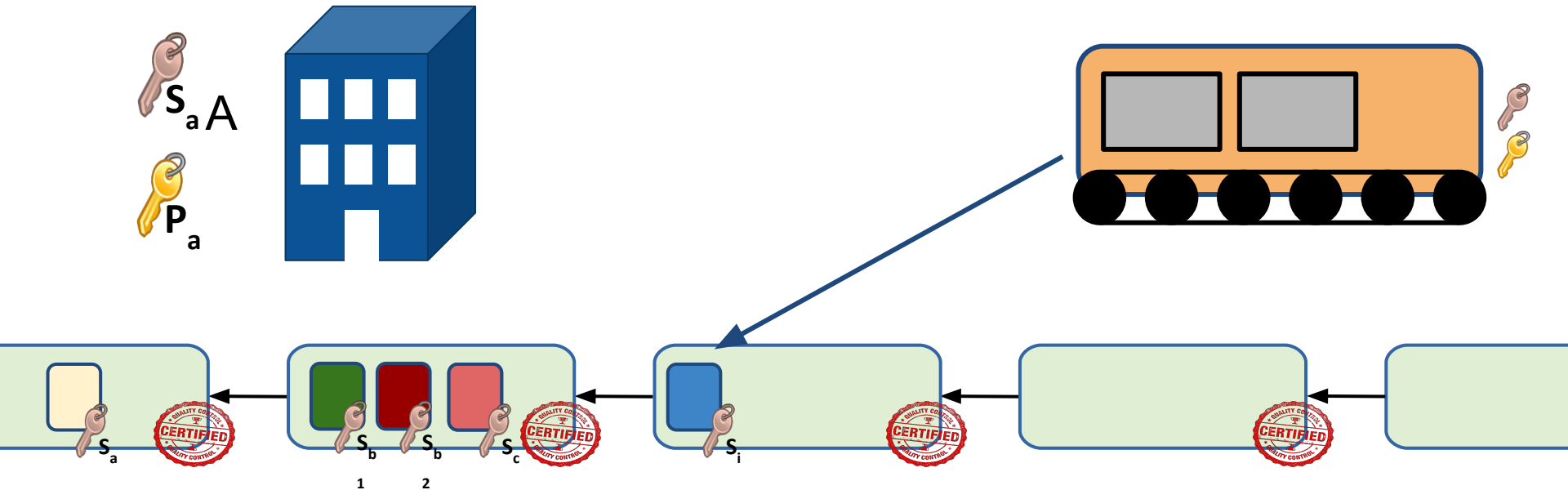
A authorizes others



Everyone writes their activities

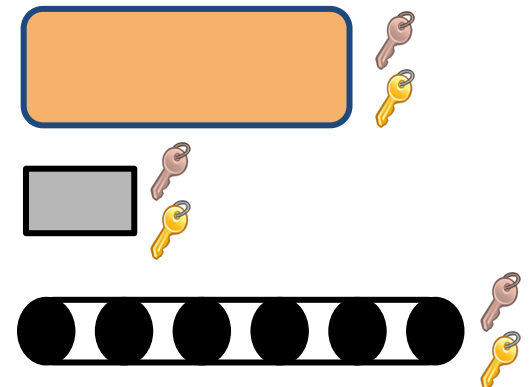


Even products can write

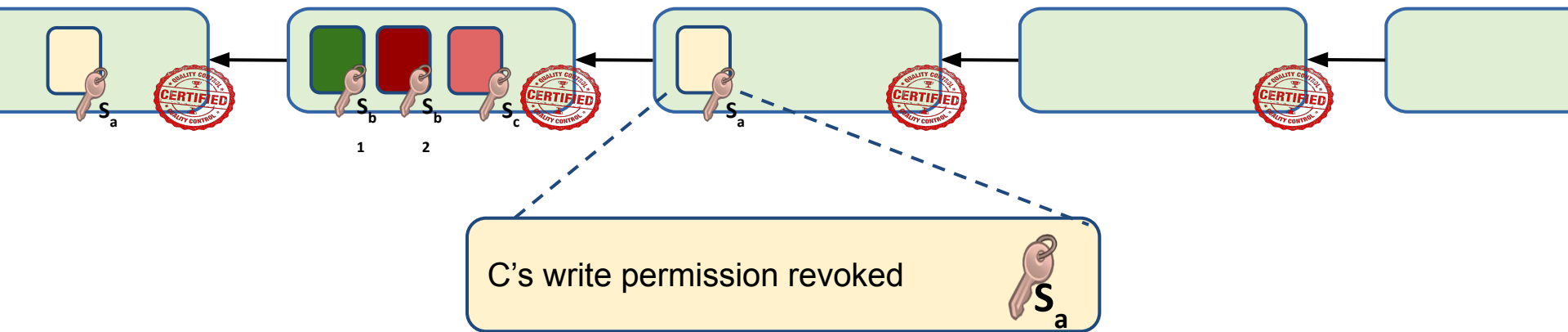
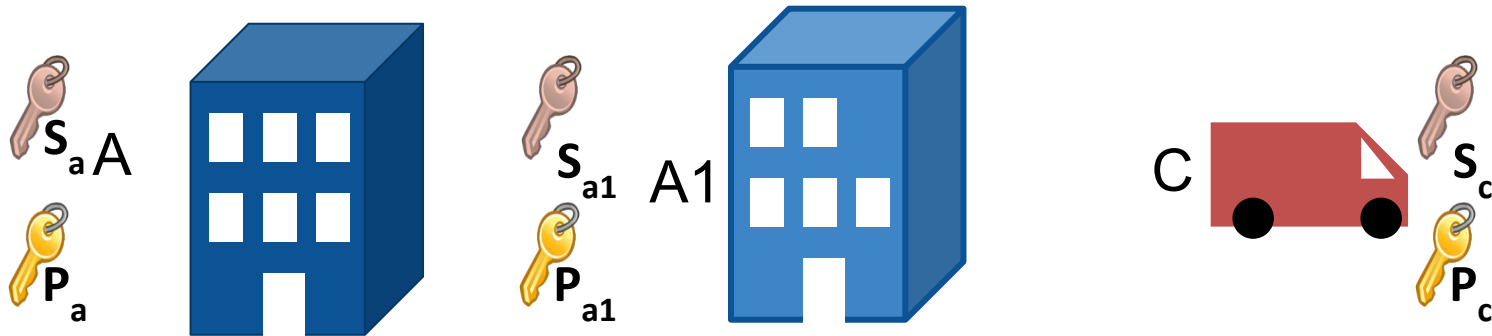


If they have sufficient “intelligence”, the products can

- register statistics, anomalies, ...
- notify maintenance activities
- ...



Revoke permissions



Every moment, A can revoke write access to someone

GDPR & Blockchain

General Data Protection and Regulation

EU n. 679/2016

- Self-executing
 - take effect without implementing a legislation
- In&Extra UE applicability
- Centralized model for handling data
- Roadmap started in 2016
- In effect since 2016
- Penalties since 2018
- Next revision 2020



GDPR and Rights for Individuals

- Right of access by the data subject (art. 15)
- Right to erasure – 'right to be forgotten' (art. 17)
- Right to rectification (art. 16)

What are the Data Subject Rights?



GDPR and Rights for Individuals

- Right of access by the data subject (art. 15)
- Right to erasure – 'right to be forgotten' (art. 17)
- Right to rectification (art. 16)

What are the Data Subject Rights?

Need to use **off-chain data**



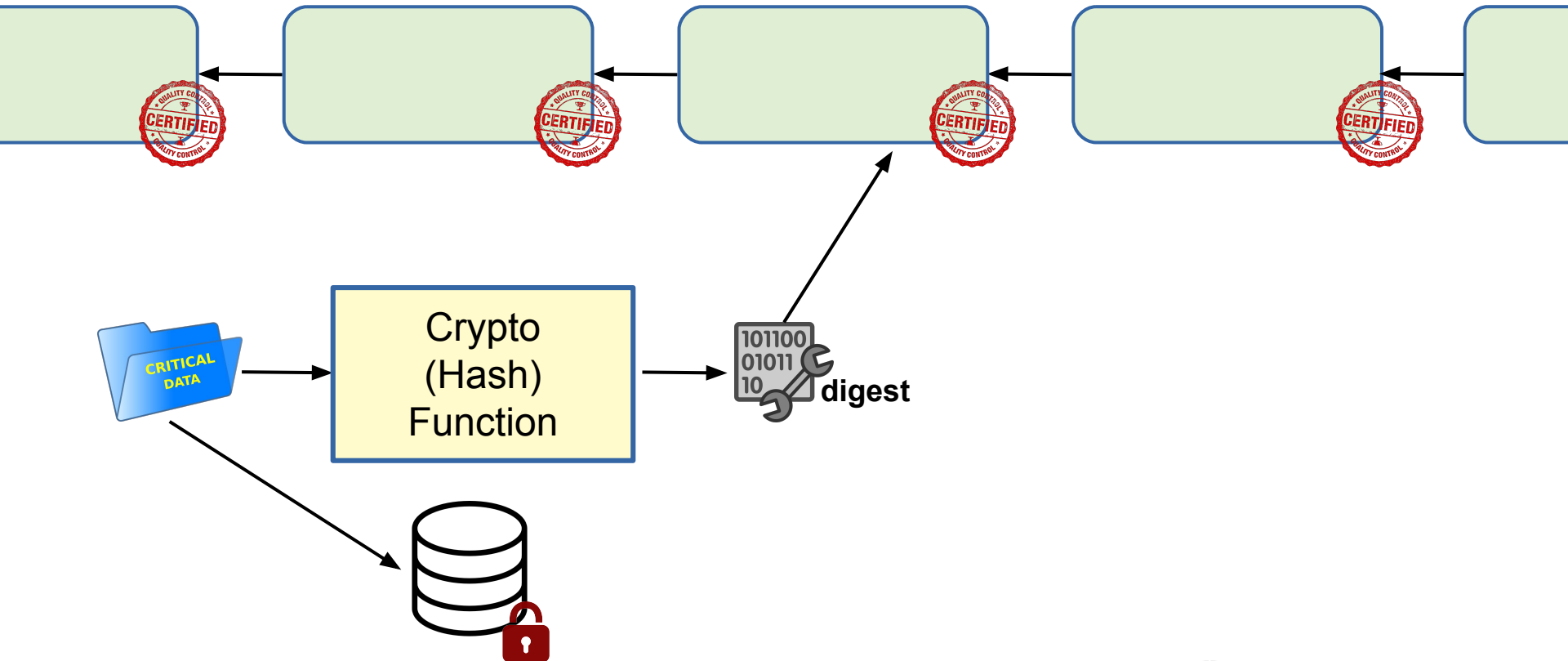
Example

- Sensitive data stored off-chain
- Smart contract controls the access to the data
 - Provide keys
 - Grants and revokes access to data
 - Records each operation → accountability



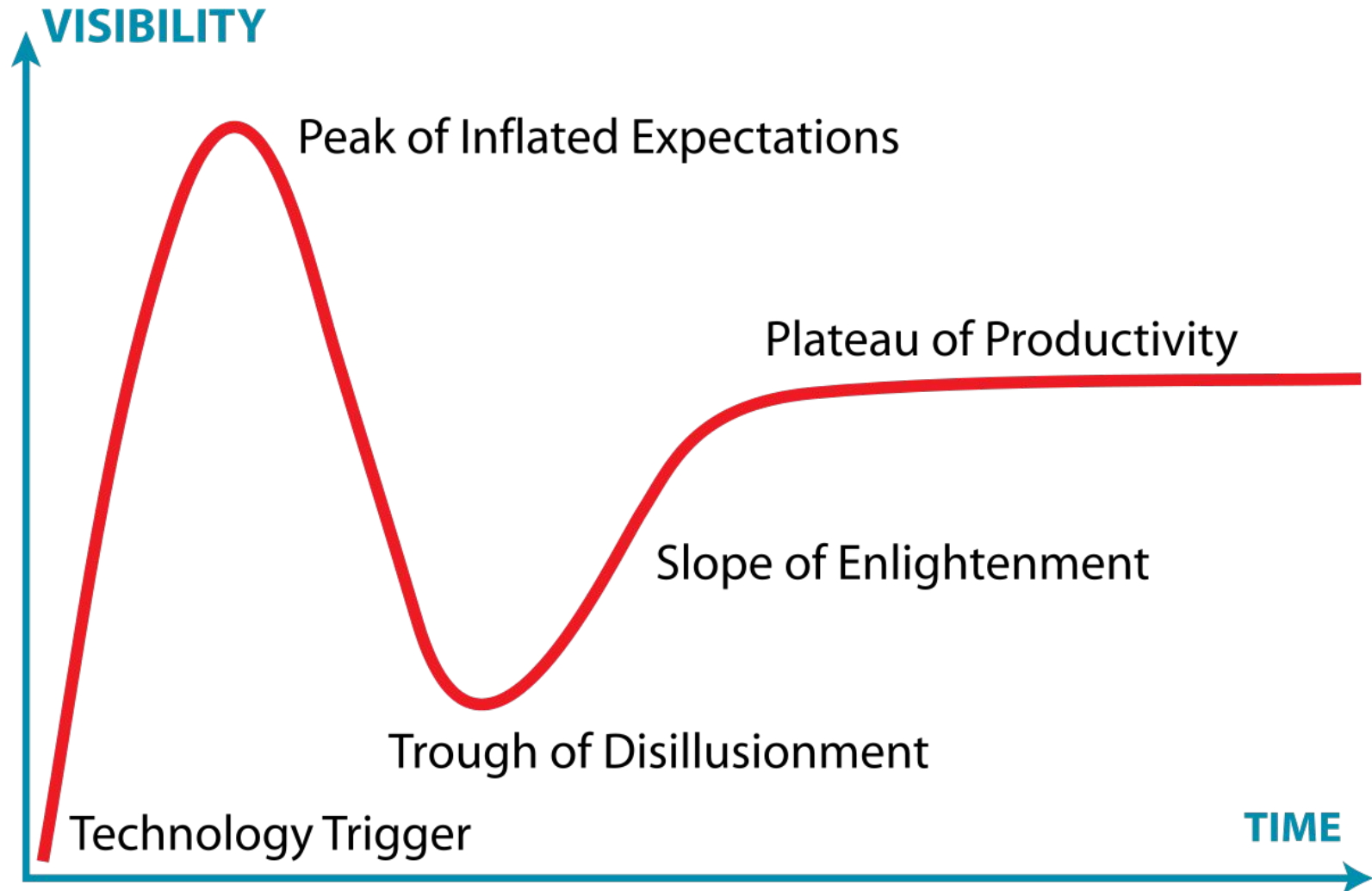
Sensitive Data

- The typical solution



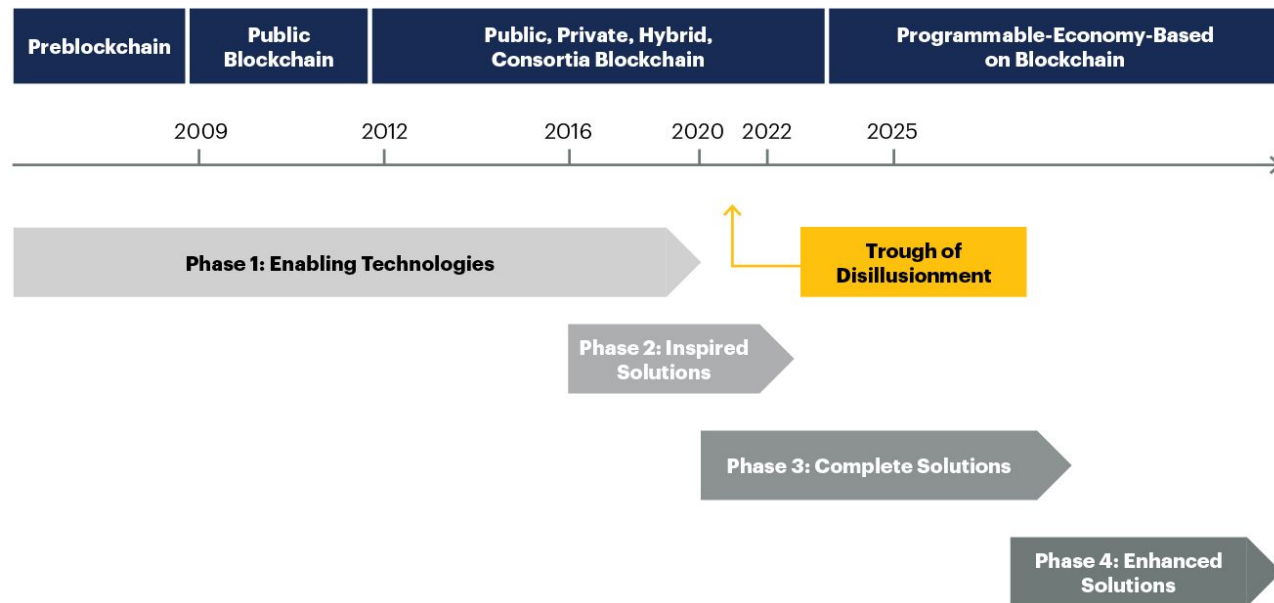
**But When I Need a
Blockchain?**

Blockchain hype-cycle



Blockchain hype-cycle

Gartner's Blockchain Spectrum



gartner.com/SmarterWithGartner

Source: Gartner
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Gartner®



DB

Private Permissioned blockchain



Permissionless blockchain



Public Permissioned blockchain



Need for a **shared** and **consistent** data store?



Private Permissioned blockchain

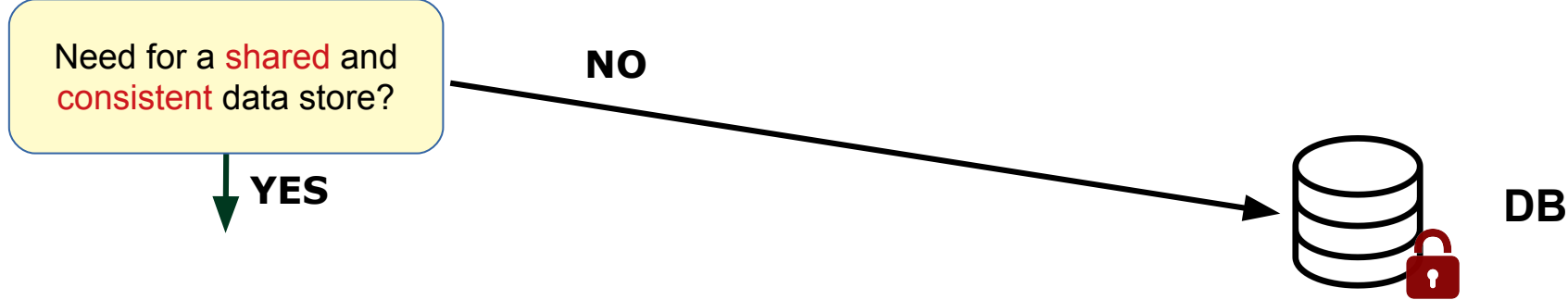


Permissionless blockchain



Public Permissioned blockchain





Private Permissioned blockchain

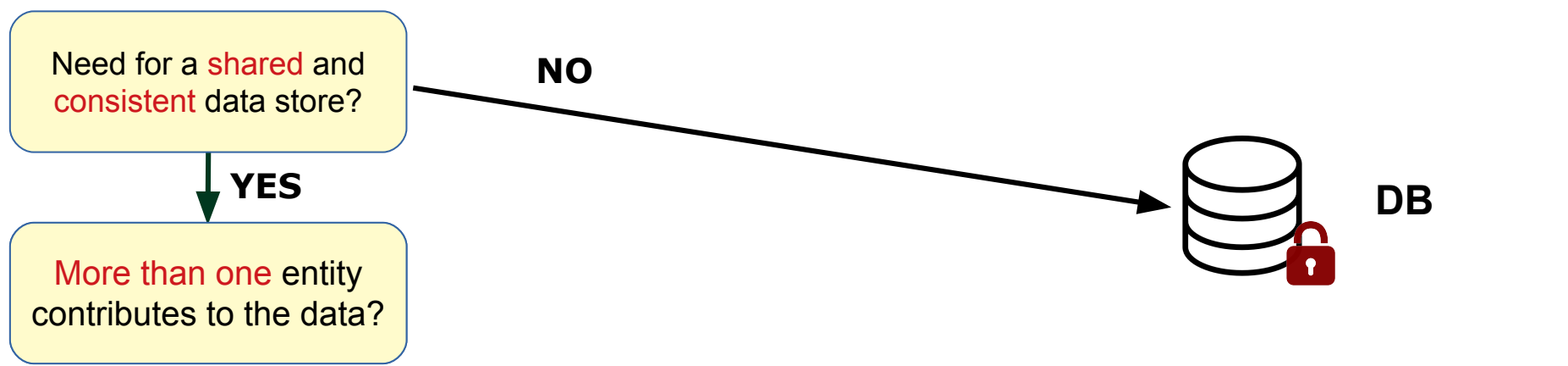


Permissionless blockchain



Public Permissioned blockchain





Private Permissioned blockchain

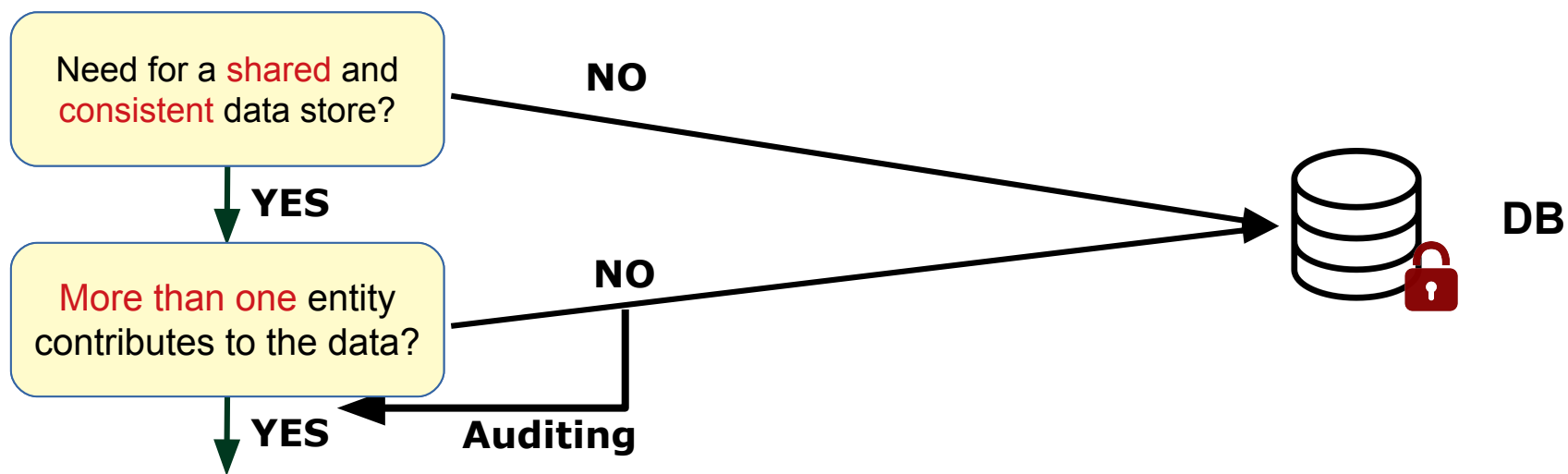


Permissionless blockchain



Public Permissioned blockchain





Private Permissioned blockchain

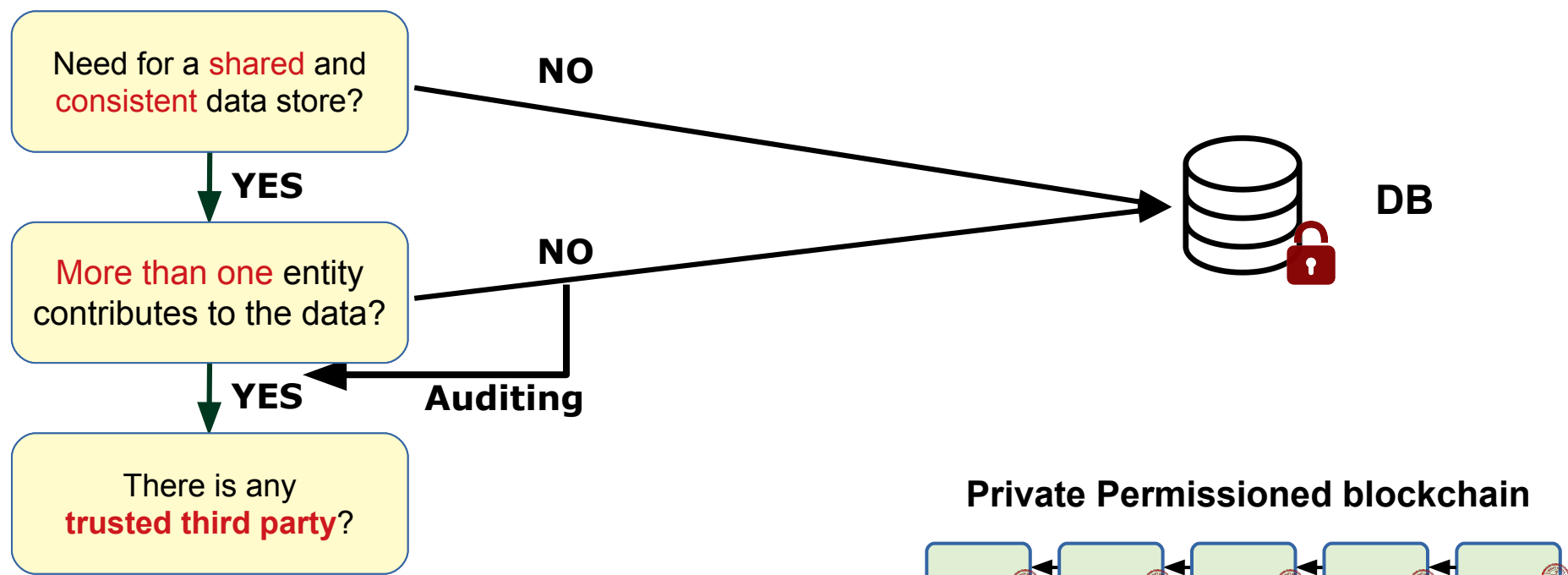


Permissionless blockchain



Public Permissioned blockchain





Private Permissioned blockchain

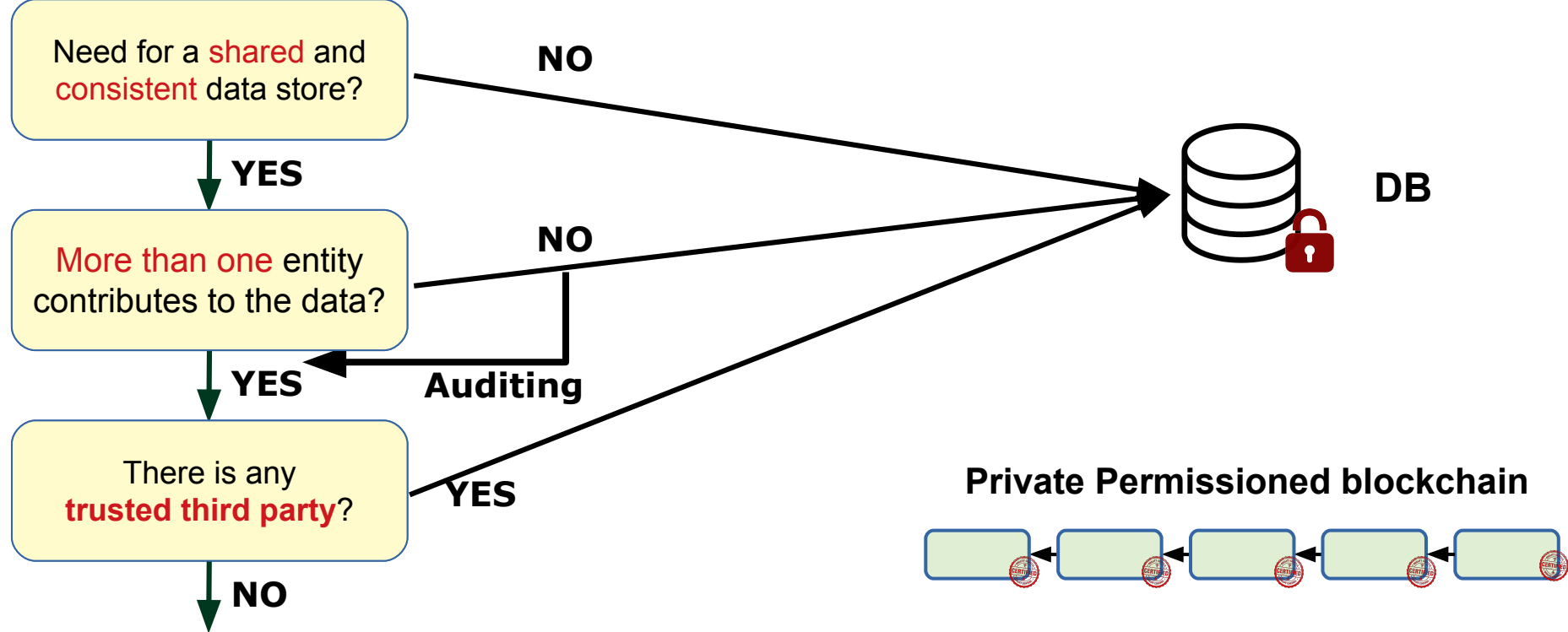


Permissionless blockchain



Public Permissioned blockchain





Private Permissioned blockchain

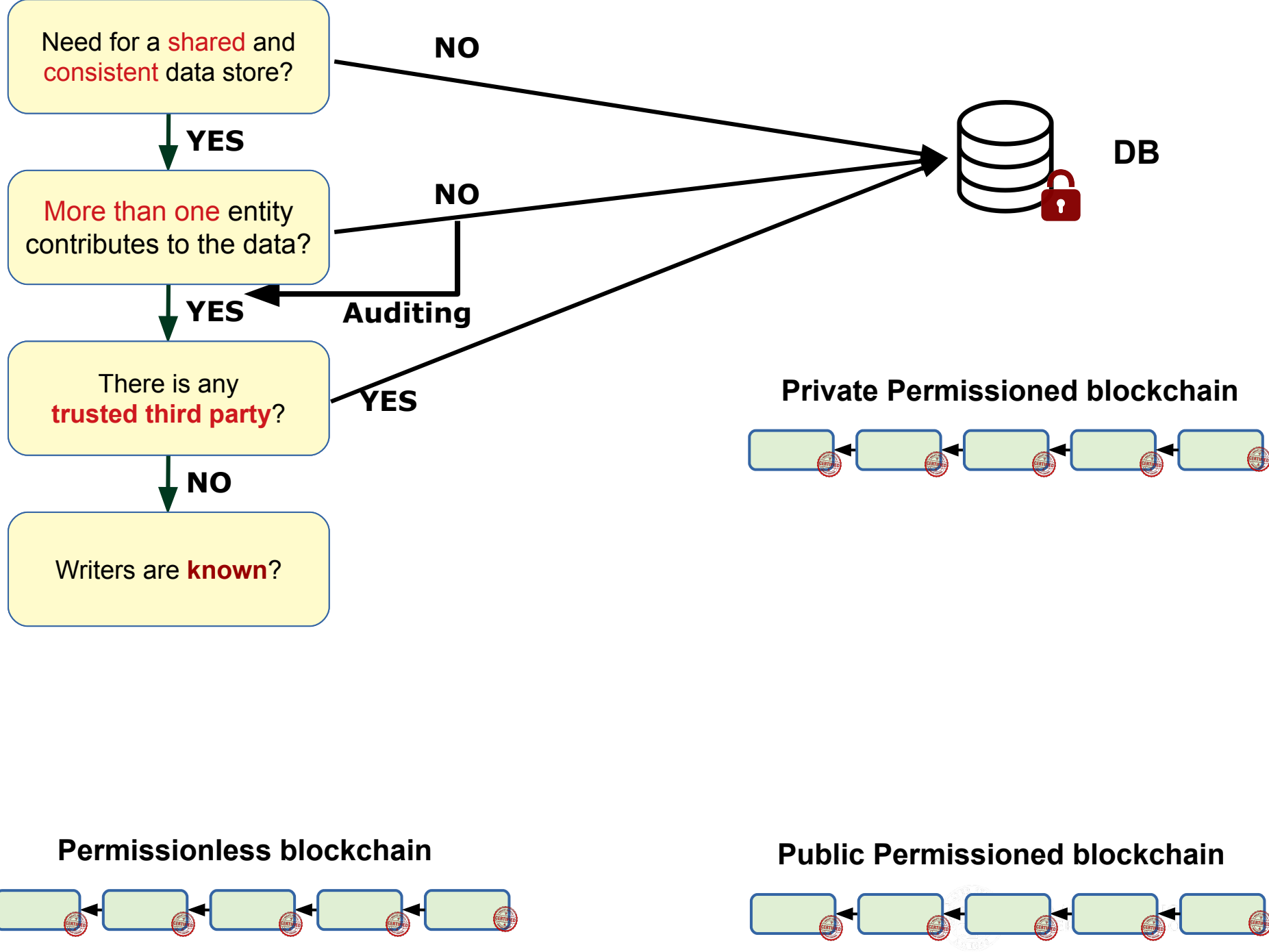


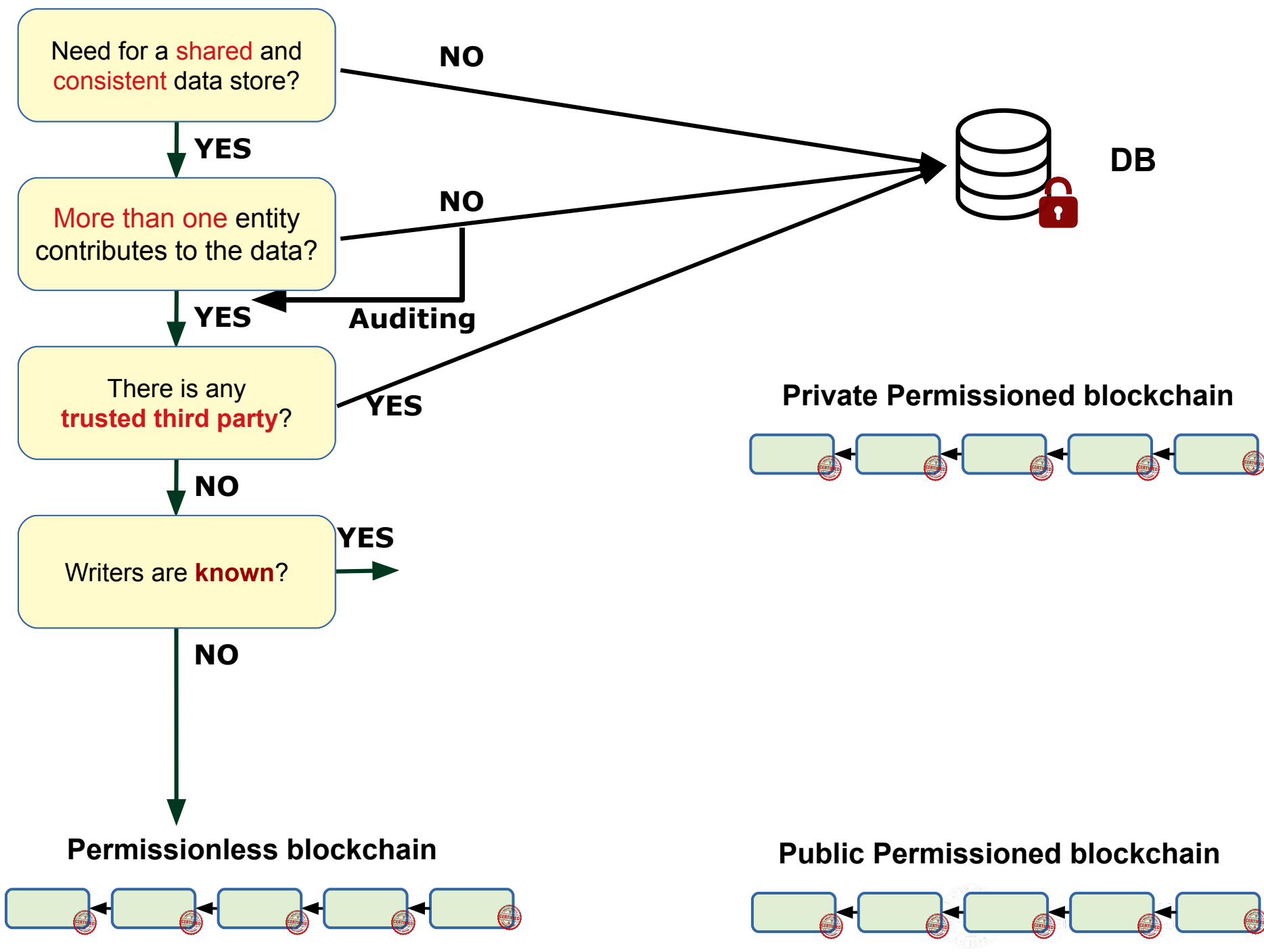
Permissionless blockchain

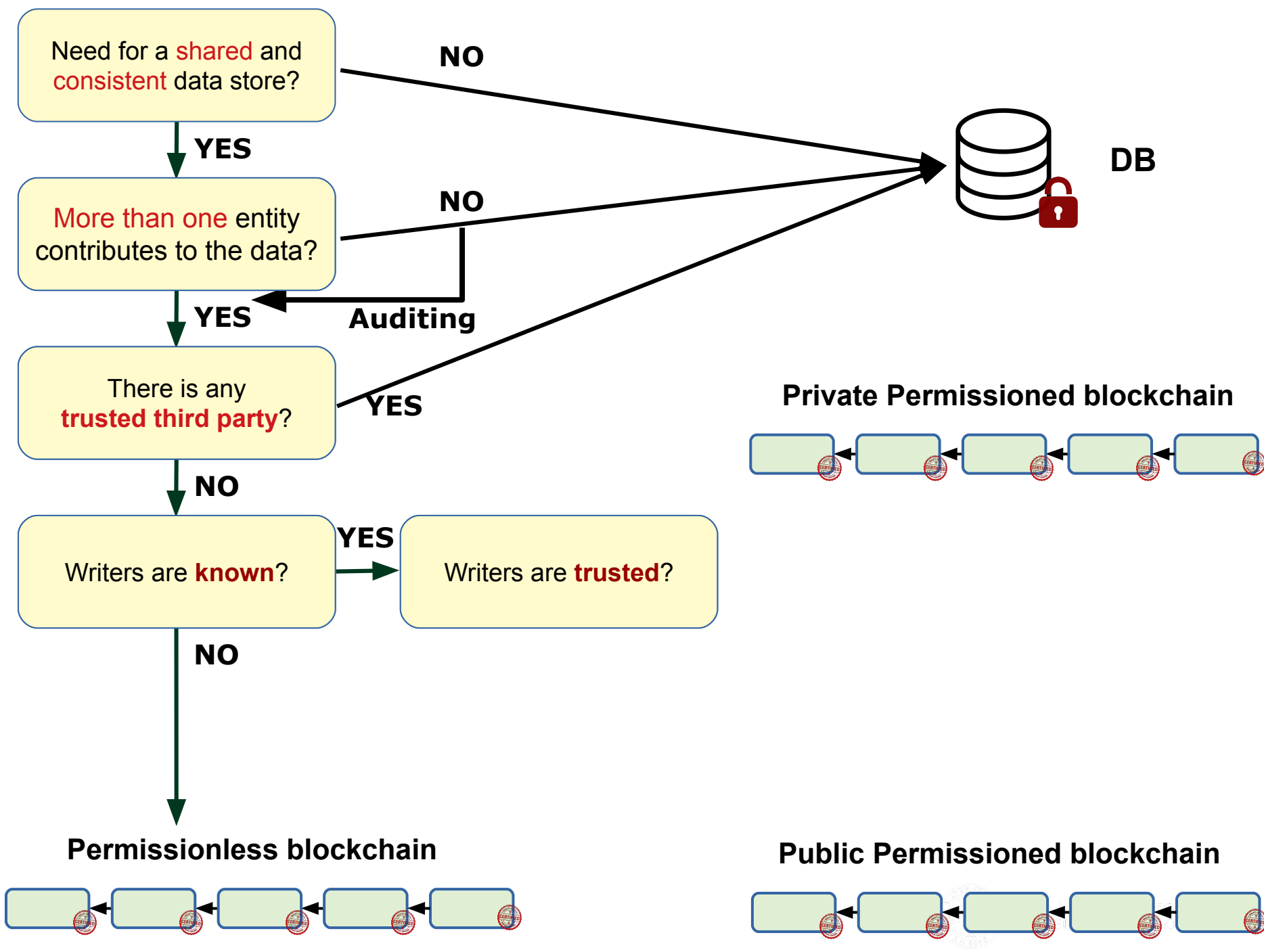


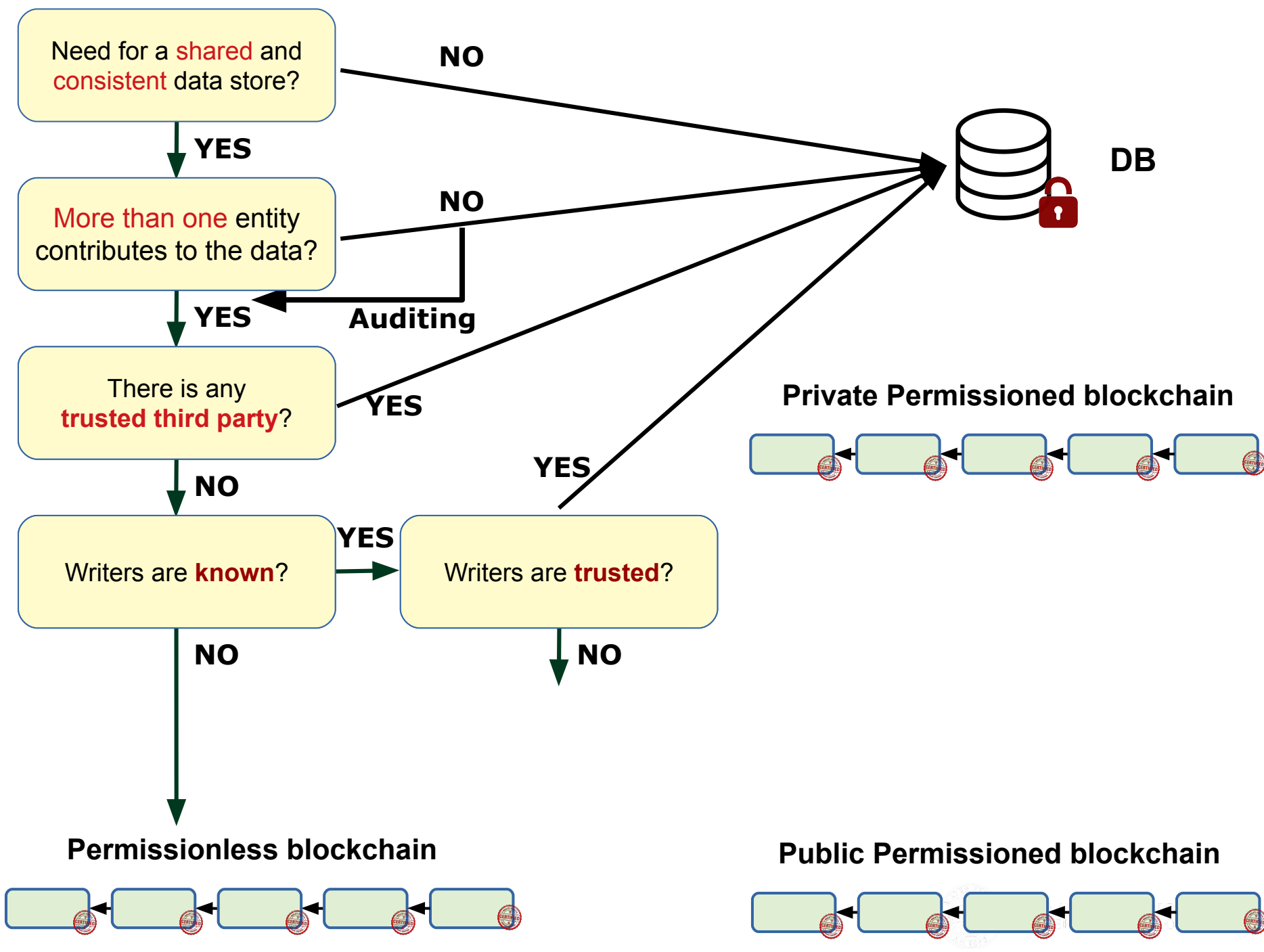
Public Permissioned blockchain

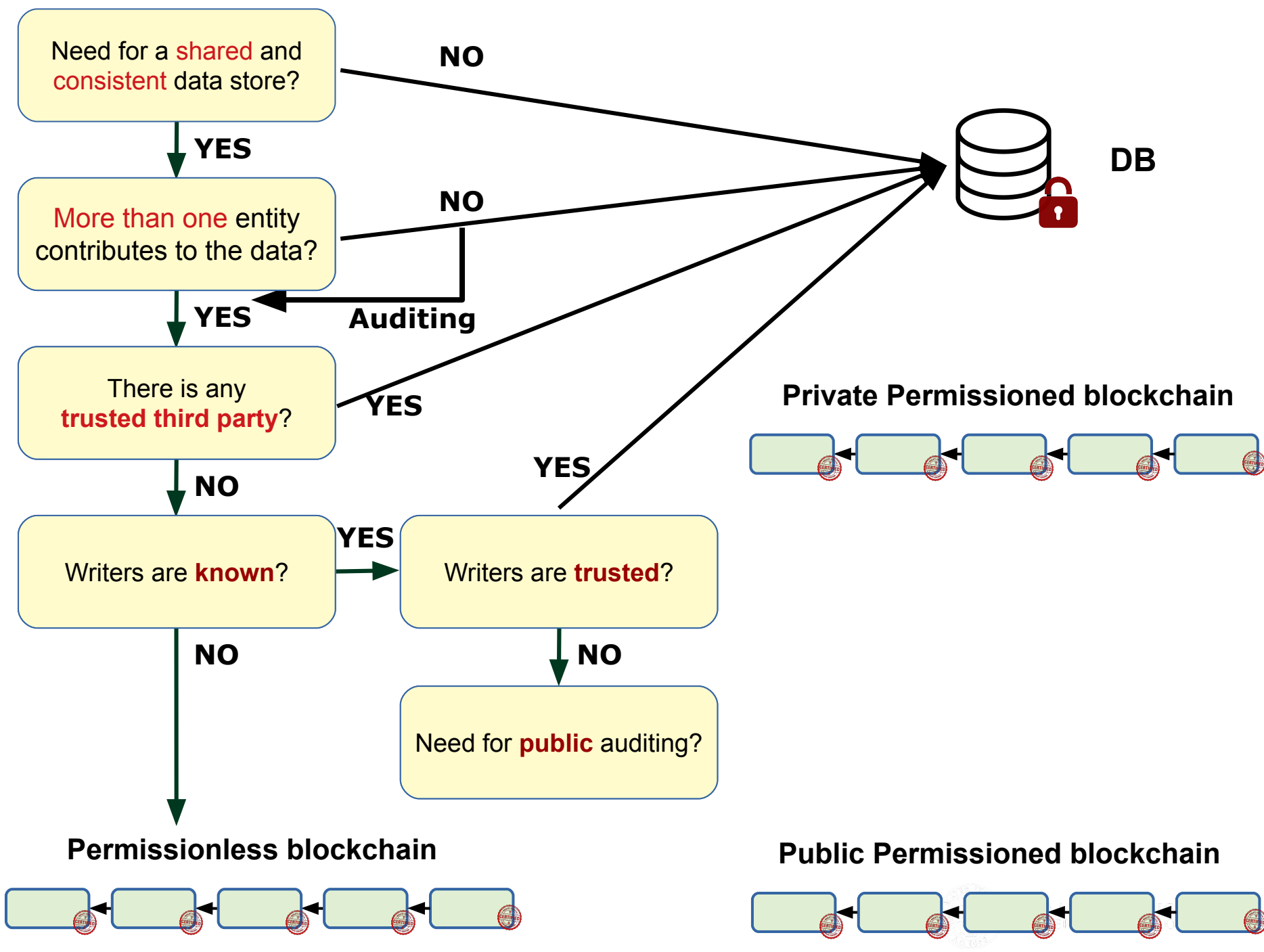


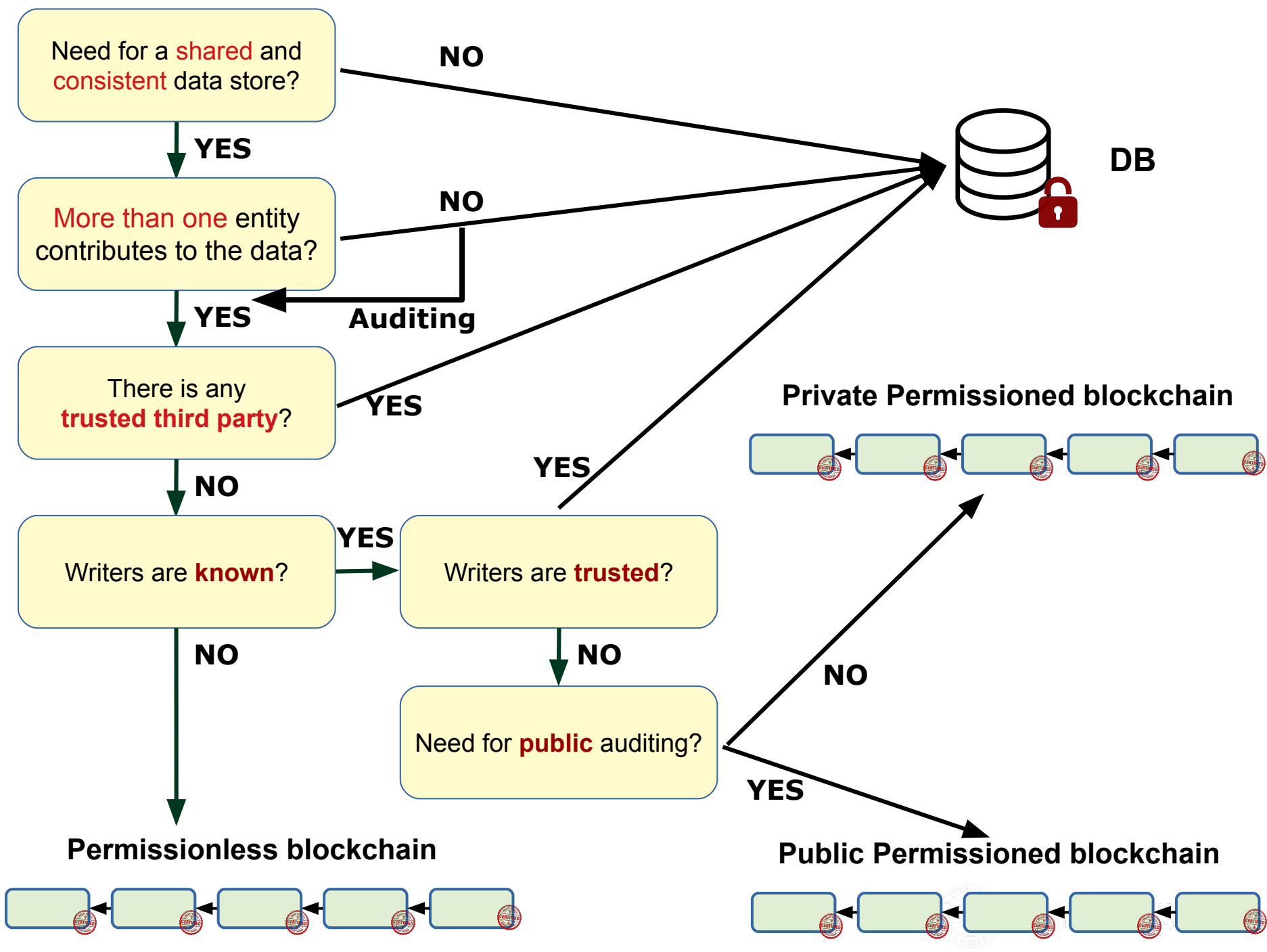












Stefano Ferretti

s.ferretti@unibo.it